机械与电子工程学院
College of Mechanical and Electronic Engineering

第十三章 能量法

Chapter13 Energy Method

刘洪萍

内容

§13-1 概述

Introduction

§13-2 杆件变形能的计算

Calculation of strain energy for various types of loadin

§13-3 变形能的普遍表达式

Reciprocal theorems

§13-7 单位荷载法•莫尔定理

Unit-load method & mohr’s theorem

§13-8 计算莫尔积分的图乘法

The meth-od of moment areas for mohr’s integration

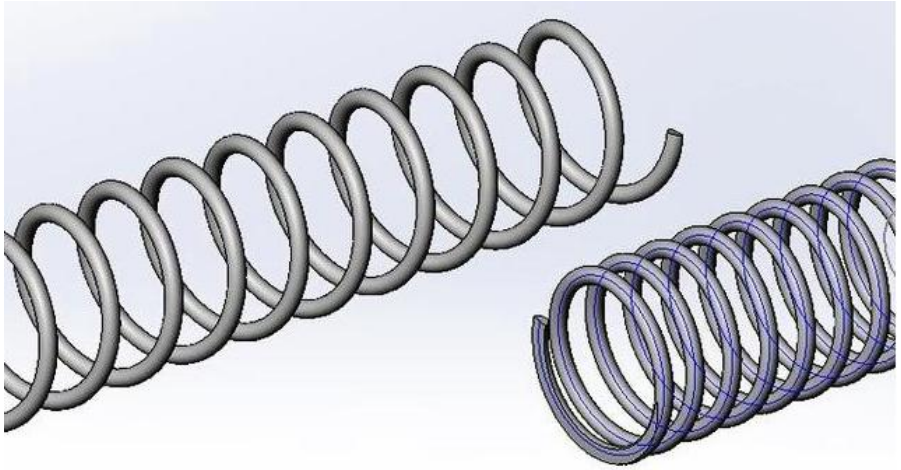
§13-1 概述（Introduction）

一、外力功（Work of the external force）

固体在外力作用下变形，引起力作用点沿力作用方向位移，外力因此而做功，则称为外力功。

二、变形能（Strain energy）

在弹性范围内，弹性体在外力作用下发生变形而在体内积蓄的能量，称为弹性变形能，简称变形能。



三、功能原理（Work-energy principle）

可变形固体在受外力作用而变形时，外力和内力均将作功。对于弹性体，不考虑其他能量的损失，外力在相应位移上作的功，在数值上就等于积蓄在物体内的应变能。

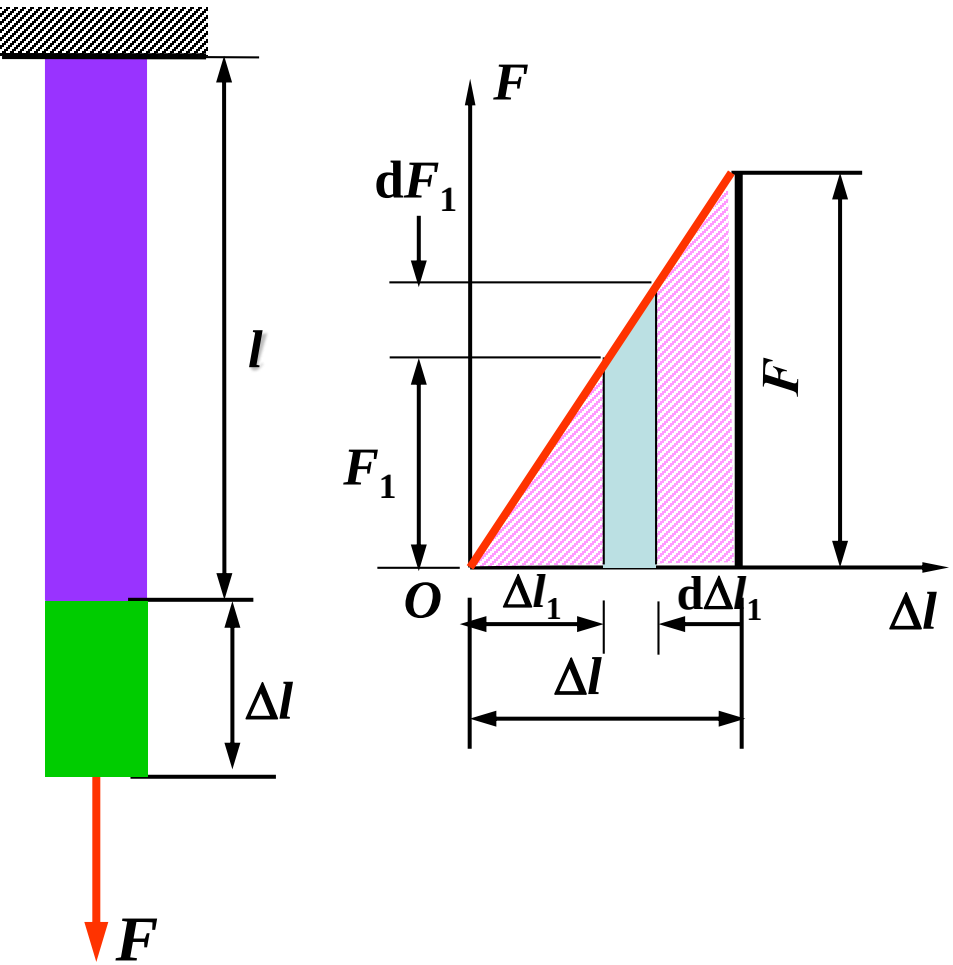
四、能量方法（Energy methods)

利用功能原理 $V_\varepsilon = W$ 来求解可变形固体的位移、变形和内力等的方法。

§ 13-2 杆件变形能的计算

(Calculation of strain energy for various types of loading)

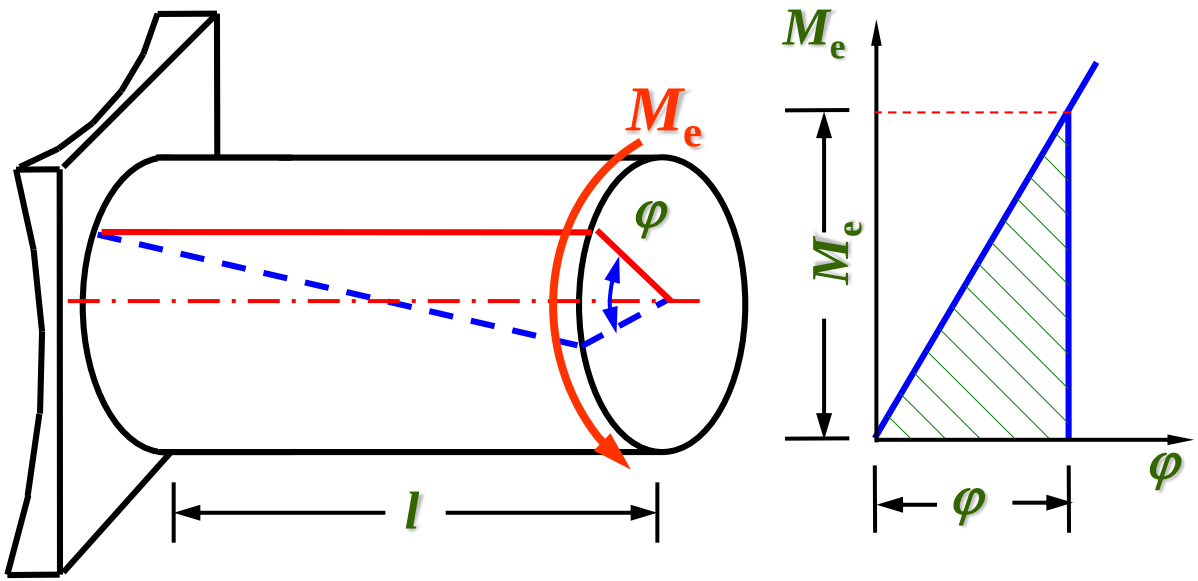
一、杆件变形能的计算（Calculation of strain energy for various types of loading）



1、轴向拉压的变形能

$$W=\frac{F}{2}\Delta l=\frac{F^2 l}{2EA}=\frac{F_N^2 l}{2EA}=V_\varepsilon$$

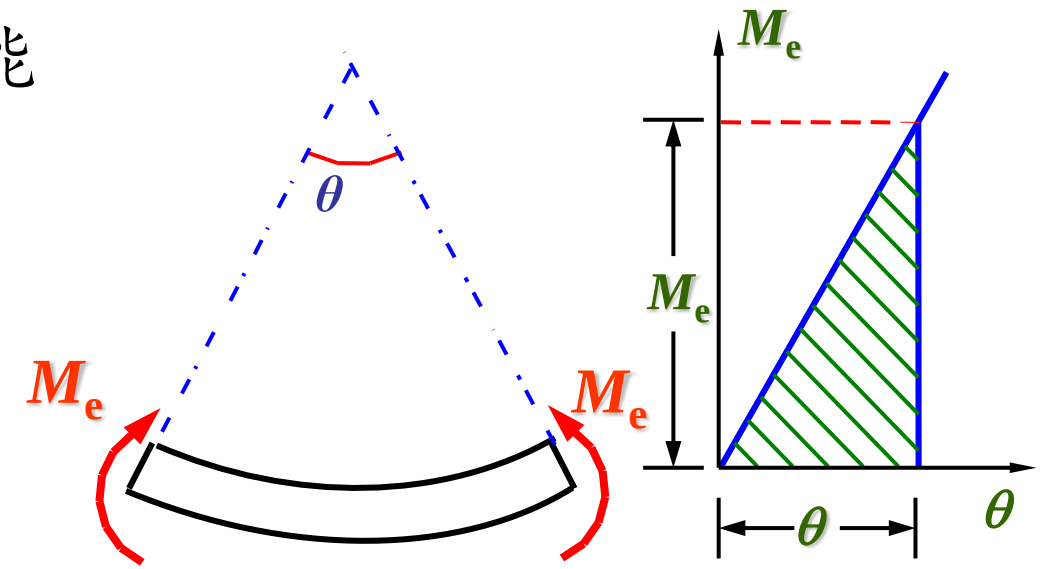
2、扭转杆内的变形能



$$V_\varepsilon = W = \frac{1}{2} M_e \cdot \Delta\varphi = \frac{1}{2} M_e \frac{M_e l}{G I_p} = \frac{M_e^2 l}{2 G I_p} = \frac{T^2 l}{2 G I_p}$$

$$V_\varepsilon = \int_l \frac{T^2(x)}{2 G I_p(x)} \mathrm{d}x \quad 或 \quad V_\varepsilon = \sum_{i=1}^{n} \frac{T_i^2 l_i}{2 G_i I_{pi}}$$

3、弯曲变形的变形能



- 纯弯曲

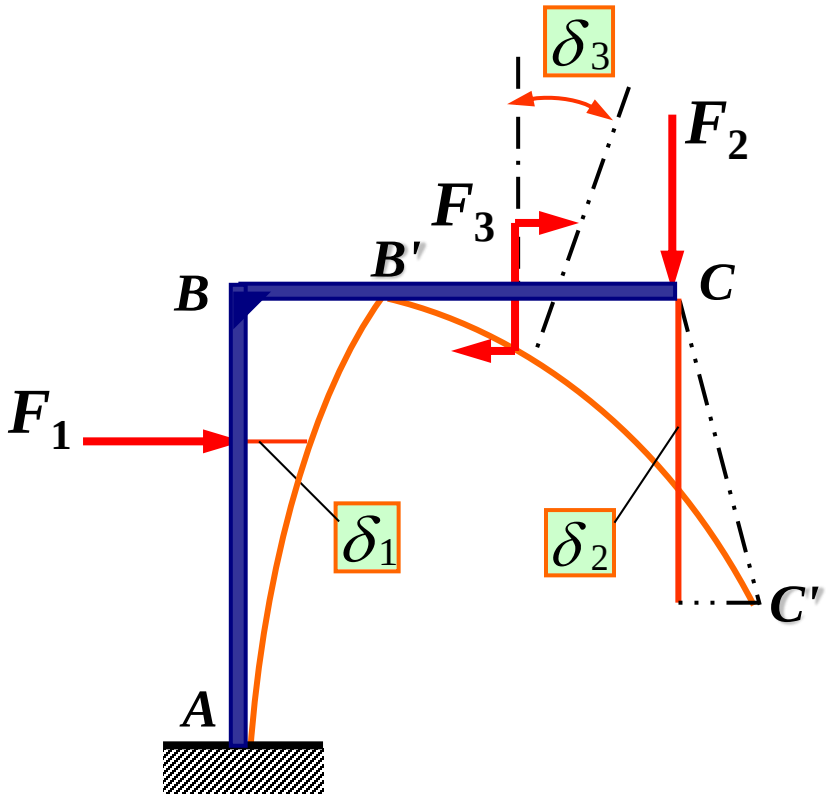
$$V_\varepsilon = W = \frac{1}{2} M_e \cdot \theta = \frac{1}{2} M_e \frac{M_e l}{EI} = \frac{M^2 l}{2EI}$$

- 横力弯曲

$$V_\varepsilon = \int_l \frac{M_e^2(x)}{2EI(x)} \mathrm{d}x$$

二、变形能的普遍表达式

（General formula for strain energy）



$$V_\varepsilon = \frac{1}{2} F\delta$$

F--广义力(包括力和力偶)

δ--广义位移

(包括线位移和角位移)

假设广义力按某一比例由零增致最后值，对应的广义位移也由零增致最后值。

对于线性结构，位移与荷载之间是线性关系，任一广义位移，例如 δ_2 可表示为

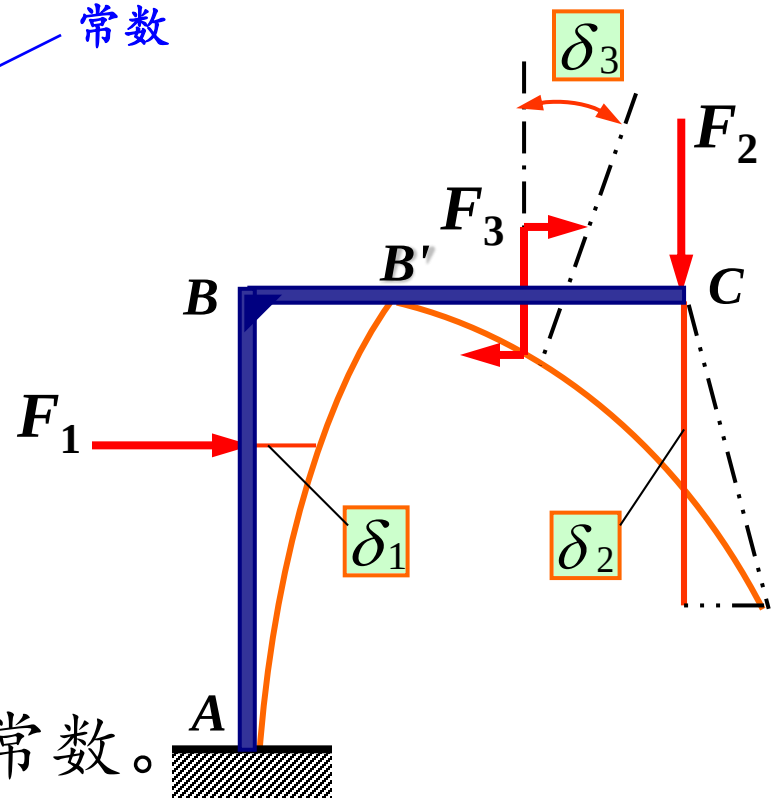
$$\delta_2 = C_1F_1 + C_2F_2 + C_3F_3$$

$$= F_2\left(\boxed{C_1\frac{F_1}{F_2} + C_2 + C_3\frac{F_3}{F_2}}\right)$$

常数

C_1、C_2、C_3 是比例常数。

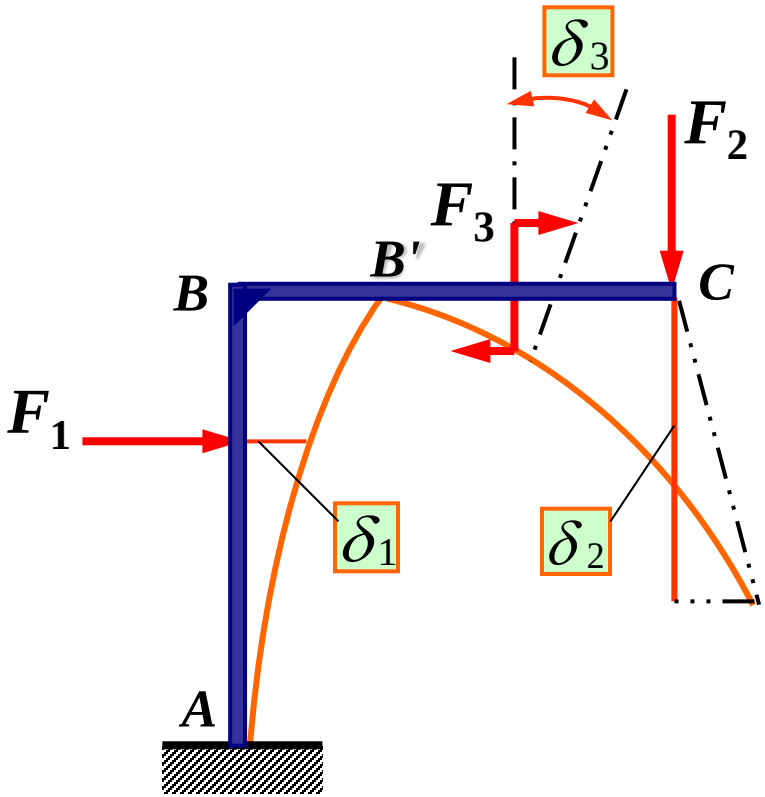
C_1F_1、C_2F_2、C_3F_3 分别表示力 F_1、F_2、F_3 在 C 点引起的广义位移。

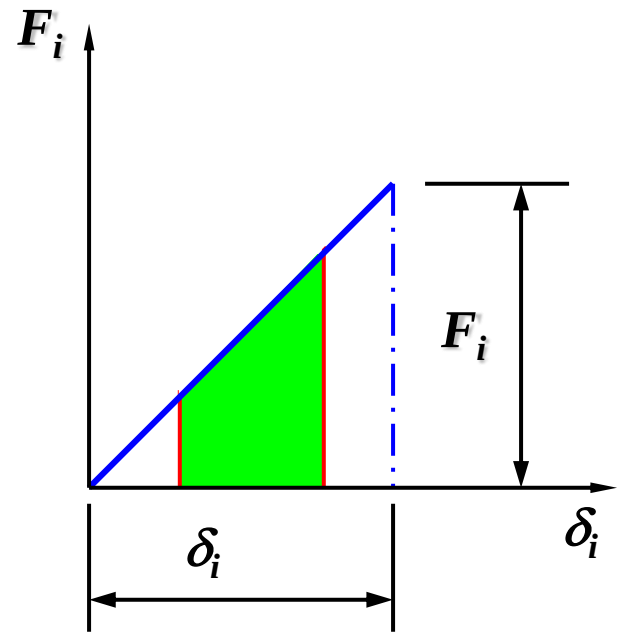


在比例加载时 F_1/F_2 和 F_3/F_2 也是常数。

δ_2 与 F_2 之间的关系是**线性**的。

同理，δ_1 与 F_1、δ_3 与 F_3 之间的关系也是**线性**的。





在整个加载过程中结构的变形能等于外力的功

$$V_\varepsilon = \frac{1}{2}(F_1\delta_1 + F_2\delta_2 + F_3\delta_3)$$

—— 克拉贝隆原理（只限于线性结构）

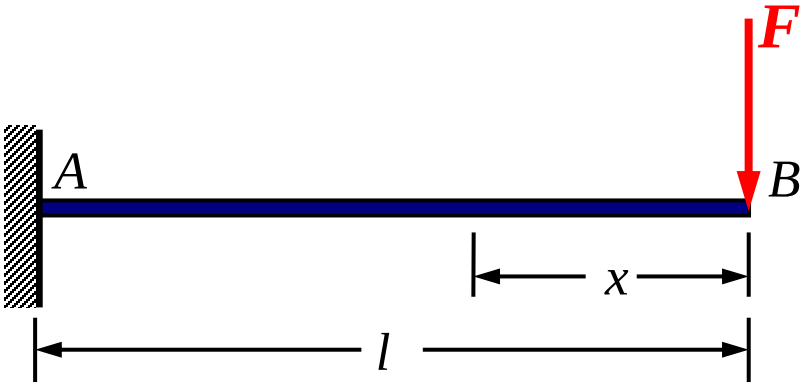
三、组合变形的变形能

截面上存在几种内力，各个内力及相应的各个位移相互独立，力独立作用原理成立，**各个内力只对其相应的位移做功**。

$$V_{\varepsilon}=\int_{l}\frac{F_{\mathrm{N}}^{2}(x)}{2EA(x)}\mathrm{d}x+\int_{l}\frac{T^{2}(x)}{2GI_{\mathrm{p}}(x)}\mathrm{d}x+\int_{l}\frac{M^{2}(x)}{2EI(x)}\mathrm{d}x$$

例题 13-1

试求图示悬臂梁的变形能，并利用功能原理求自由端B的挠度。



解： $M(x) = -F \cdot x$

$$V_\varepsilon = \int_l \frac{M^2(x)}{2EI}\mathrm{d}x = \int_0^l \frac{(Fx)^2}{2EI}\mathrm{d}x = \frac{F^2 l^3}{6EI}$$

$$W = \frac{1}{2}F \cdot w_B$$

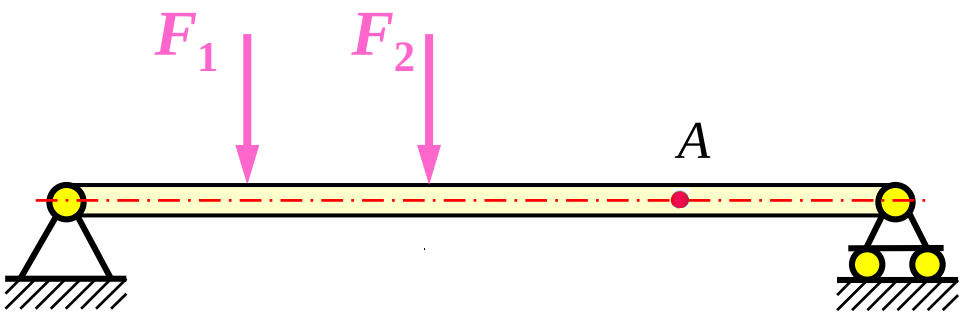
由$V_\varepsilon = W$得 $$w_B = \frac{Fl^3}{3EI}$$

§13-7 单位荷载法 • 莫尔定理

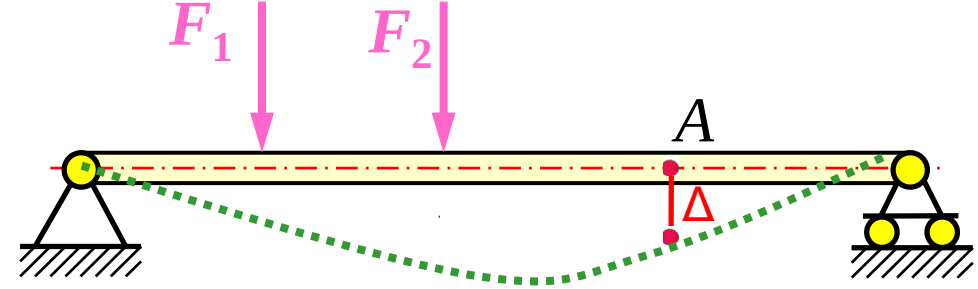
（Unit-load method & mohr’s theorem）

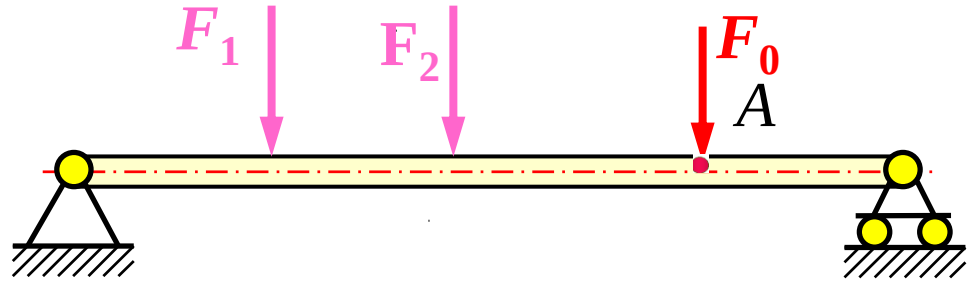
一、莫尔定理的推导（Derivation of mohr’s theorem）

求任意点A的位移w_A



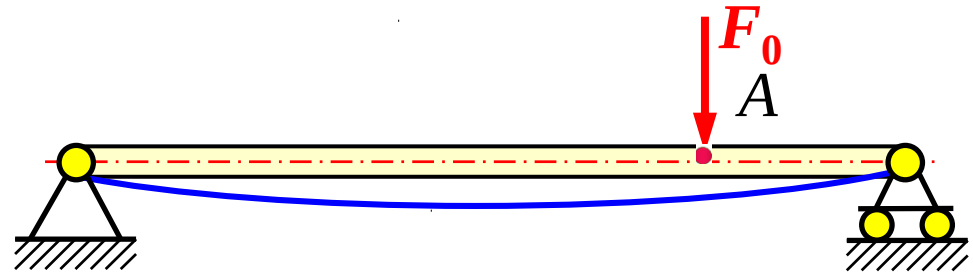
（1）先在A点作用单位力F_0， F_0=1，再作用F_1、F_2



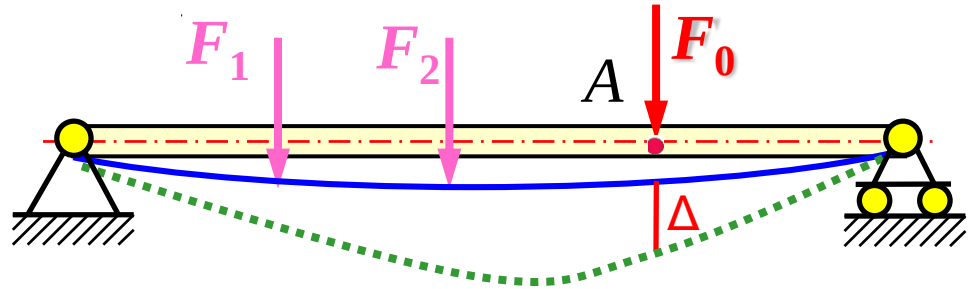


变形能为

$$V_\varepsilon = \int_l \frac{M^2(x)}{2EI}\mathrm{d}x$$



$$\overline{V}_\varepsilon = \int_l \frac{\overline{M}^2(x)}{2EI}\mathrm{d}x$$



$$V_{\varepsilon 1} = V_\varepsilon + \overline{V}_\varepsilon + 1 \times \Delta$$

（2）三个力同时作用时

任意截面的弯矩： $M(x)+\bar{M}(x)$

变形能： $V_{\varepsilon 2}=\int_l \frac{[M(x)+\bar{M}(x)]^2}{2EI}\mathrm{d}x$

$$V_{\varepsilon 2}=V_{\varepsilon 1} \Longrightarrow V_{\varepsilon}+\bar{V}_{\varepsilon}+1\cdot\Delta=\int_l \frac{[M(x)+\bar{M}(x)]^2}{2EI}\mathrm{d}x$$

$$V_{\varepsilon}+\bar{V}_{\varepsilon}+1\cdot\Delta=\int_l \frac{[M(x)+\bar{M}(x)]^2}{2EI}\mathrm{d}x$$

$$=\int_l \frac{M^2(x)}{2EI}\mathrm{d}x+\int_l \frac{\bar{M}^2(x)}{2EI}\mathrm{d}x+\int_l \frac{M(x)\bar{M}(x)}{EI}\mathrm{d}x$$

$$\boxed{\Delta = \int_l \frac{M(x)\overline{M}(x)}{EI}\mathrm{d}x}$$

应变能的特点：

（1） 应变能只与外力的最终值有关，而与加载过程和加载次序无关。

（2）产生同一种基本变形的一组外力在杆上所产生的应变能不等于各力分别作用时产生的变形能之和。

$$w_A = \int_l \frac{M(x)\overline{M}(x)}{EI}\mathrm{d}x$$

$$\theta = \int_l \frac{M(x)\overline{M}(x)}{EI}\mathrm{d}x$$

$$\Delta = \sum_{i=1}^{n} \frac{\overline{F}_{\mathrm{N}i} F_{\mathrm{N}i} l_i}{EA}$$

二、普遍形式的莫尔定理

（General formula for mohr’s theorem）

$$\Delta = \int_l \frac{F_N(x)\overline{F}_N(x)}{EA}\mathrm{d}x + \int_l \frac{T(x)\overline{T}(x)}{GI_p}\mathrm{d}x + \int_l \frac{M(x)\overline{M}(x)}{EI}\mathrm{d}x$$

一个线位移	一个角位移	相对线位移	相对角位移
⇓	⇓	⇓	⇓
一个力	一个力偶	一对力	一对力偶

三、使用莫尔定理的注意事项

（1）M（x）：结构在原载荷下的内力；

（2）$\overline{M}$：去掉主动力，在所求广义位移点，沿所求广义位移的方向加广义单位力时，结构产生的内力；

（３）位移看成广义位移，把单位力看成与广义位移相对应的广义力；

所加广义单位力与所求广义位移之积，必须为功的量纲；

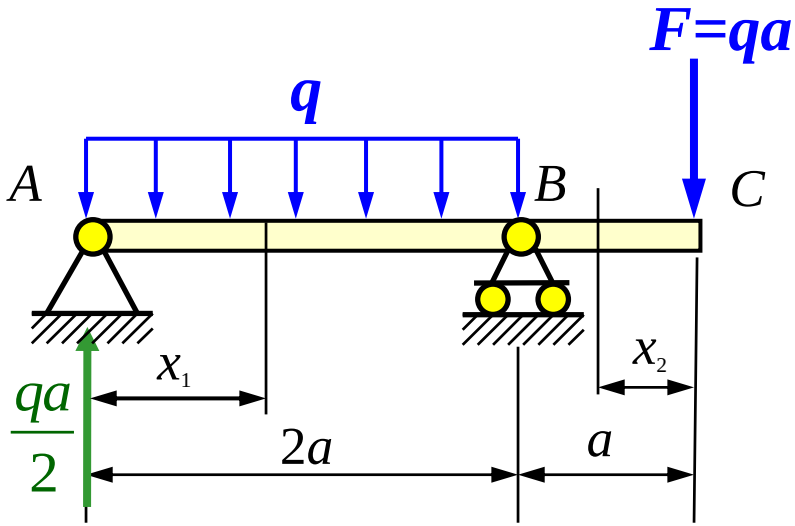
（4）$\overline{M}$（x）与M（x）的坐标系必须一致，每段杆的坐标系可自由建立；

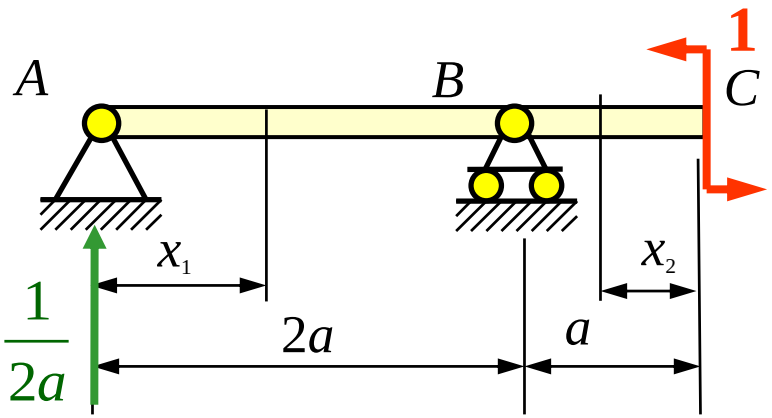
（5）莫尔积分必须遍及整个结构。

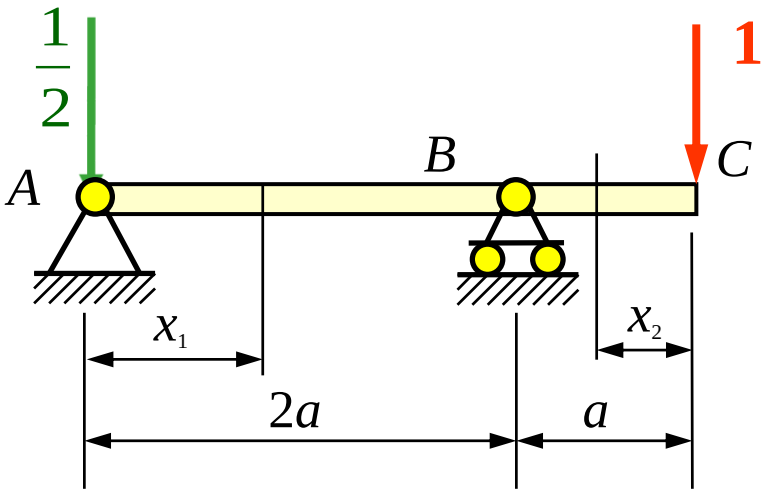
例题 13-2

图示外伸梁，其抗弯刚度为EI。

用单位载荷法求C点的挠度和转角。

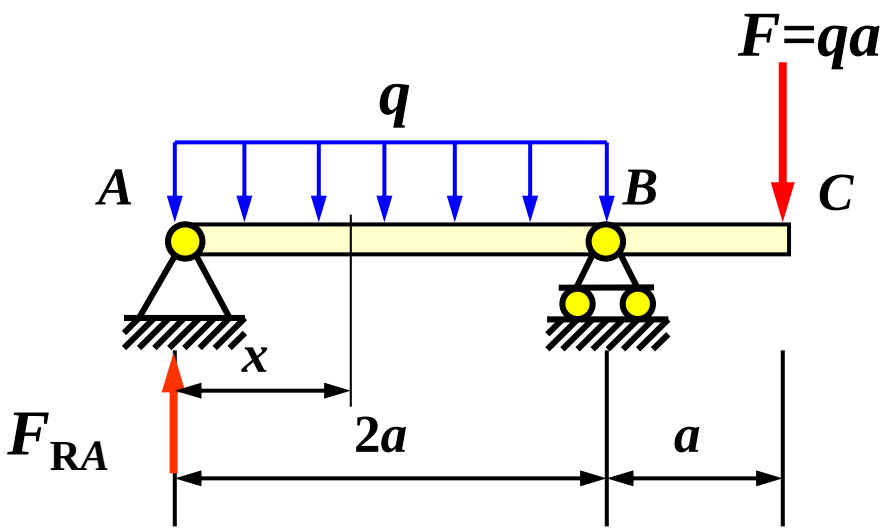


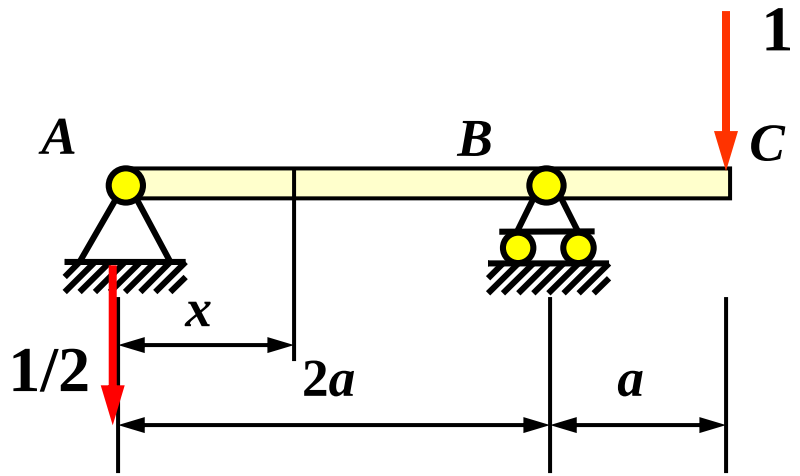




$$w_C = \frac{2qa^4}{3EI}(\downarrow)$$

$$\theta_C = -\frac{5qa^3}{6EI}(\circlearrowleft)$$

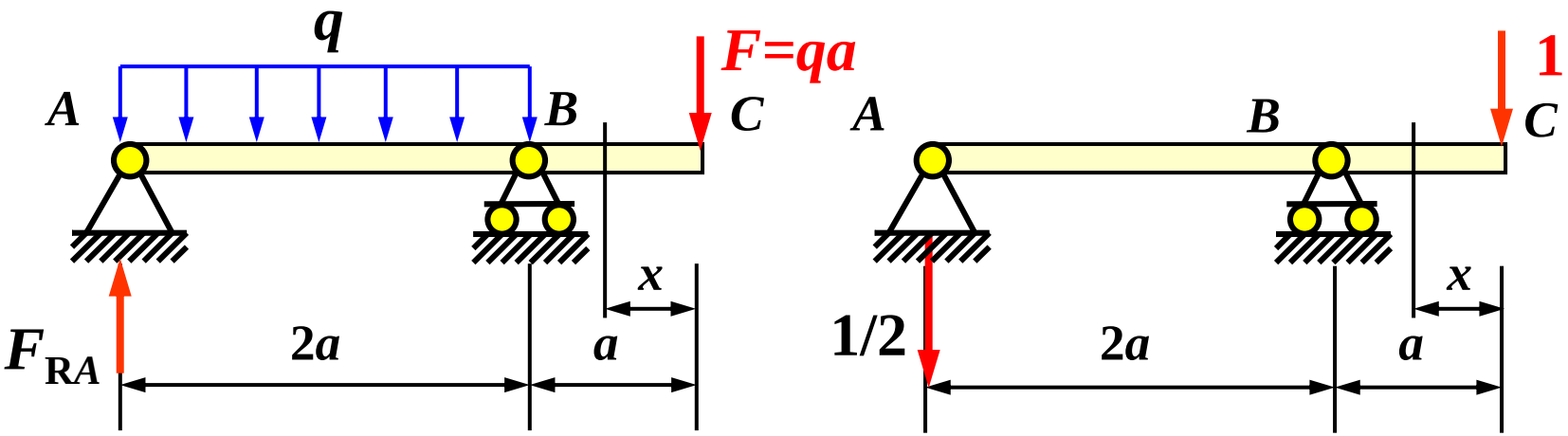




解： $F_{RA} = \dfrac{qa}{2}$

（1）求截面的挠度（在C 处加一单位力“1”）

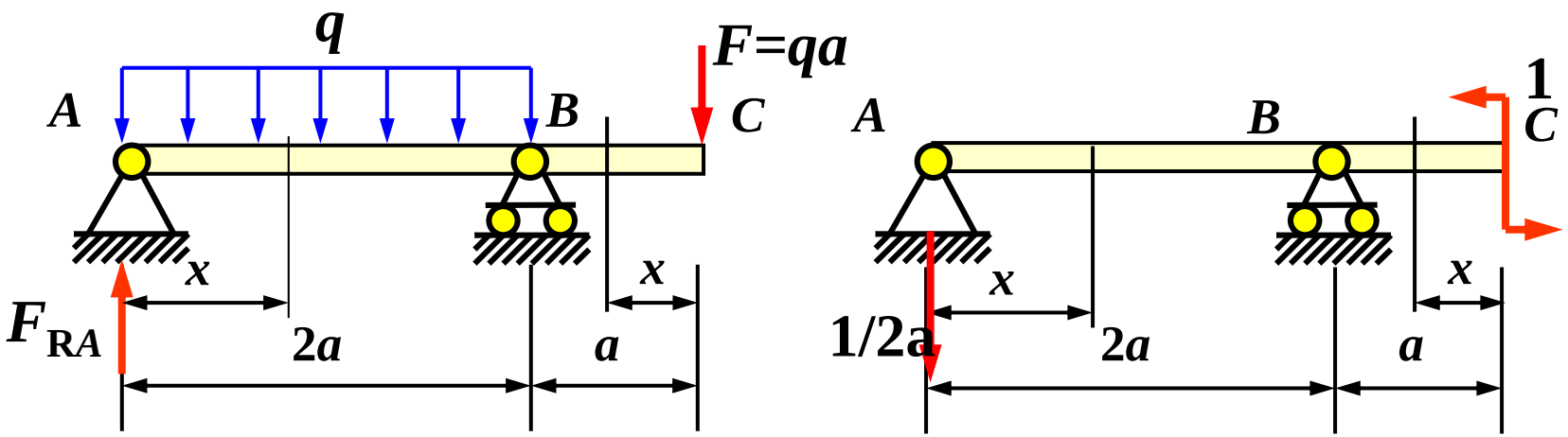
AB: $M(x) = \dfrac{qa}{2}x - \dfrac{qx^2}{2}$ $\quad M(x) = -\dfrac{x}{2}$



BC: $M(x) = -qa \cdot x \qquad \overline{M}(x) = -x$

$$w_C = \frac{1}{EI}\left[\int_0^{2a}\left(\frac{qa}{2}x - \frac{qx^2}{2}\right)\left(-\frac{x}{2}\right)\mathrm{d}x + \int_0^{a}(-qax)(-x)\mathrm{d}x\right]$$

$$= \frac{2qa^4}{3EI} \quad (\downarrow)$$



（2）求C 截面的转角（在C处加一单位力偶）

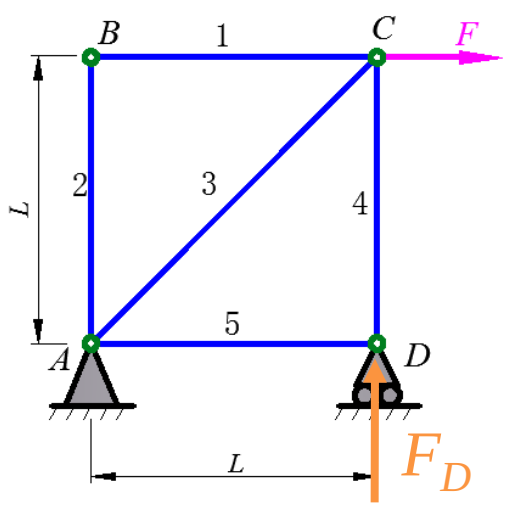
AB: $M(x)=\dfrac{qa}{2}x-\dfrac{qx^2}{2}$ $\quad \overline{M}(x)=\dfrac{x}{2a}$

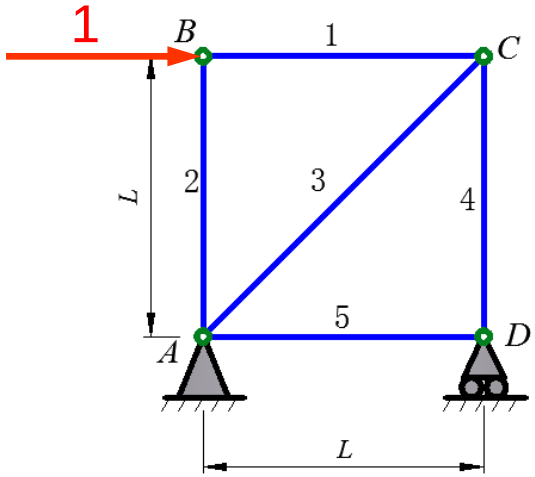
BC: $M(x)=-qa\cdot x$ $\quad \overline{M}(x)=1$

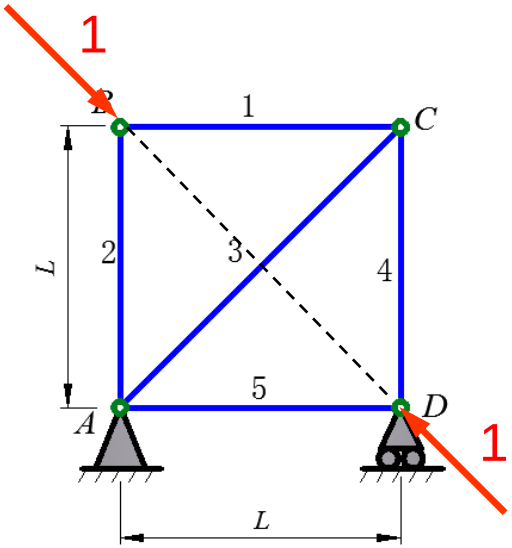
$$\theta_C=\frac{1}{EI}\left[\int_0^{2a}\left(\frac{qa}{2}x-\frac{qx^2}{2}\right)\left(\frac{x}{2a}\right)\mathrm{d}x+\int_0^{a}(-qax)(1)\mathrm{d}x\right]$$

$$=-\frac{5qa^3}{6EI}$$ （↻）

例题 13-3 桁架中各杆的EA相同，在F的作用下求B点的水平位移和B、D间的相对位移。







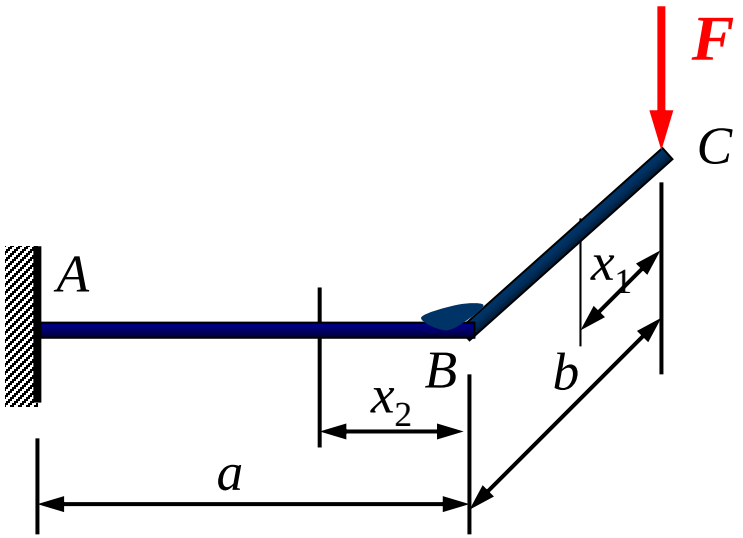
		1	2	3	4	5
原力	F_{Ni}					
水平单位载荷	F'_{Ni}					
相对单位载荷	F''_{Ni}					
	L_i	L	L	$\sqrt{2}L$	L	L
	$F_{Ni}F'_{Ni}L_i$					
	$F_{Ni}F''_{Ni}L_i$					

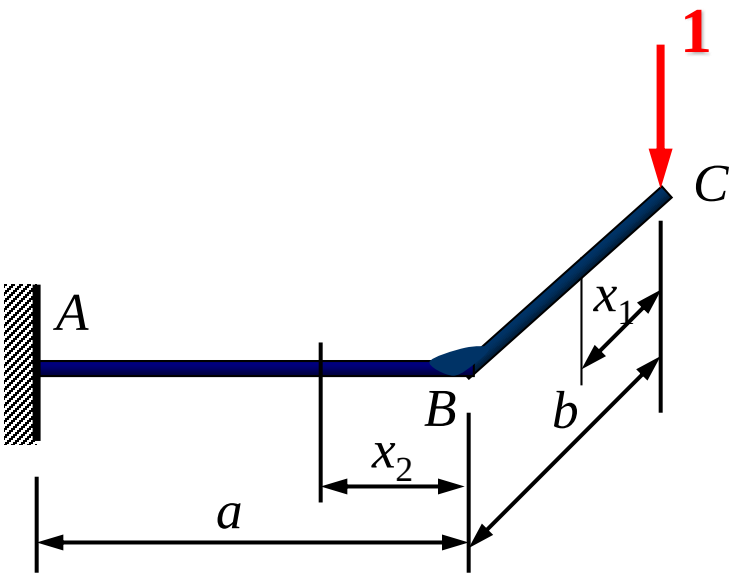
$$\Delta B_H = \frac{FL}{EA}\left(2\sqrt{2}+1\right)(\rightarrow)$$

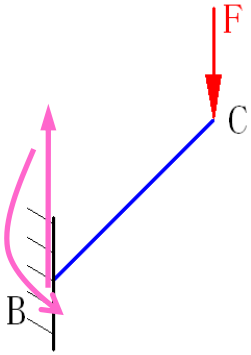
$$\delta_{BD} = \left(2+\sqrt{2}/2\right)\frac{\mathrm{FL}}{\mathrm{EA}}\text{（靠近）}$$

例题 13-4

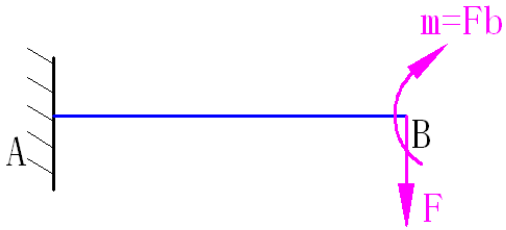
图示为一水平面内的曲杆，B 处为一刚性节点，$\angle ABC=90°$ 在 C 处承受竖直力F，设两杆的抗弯刚度和抗扭刚度分别是 EI 和 GI_p。求C点竖向的位移。

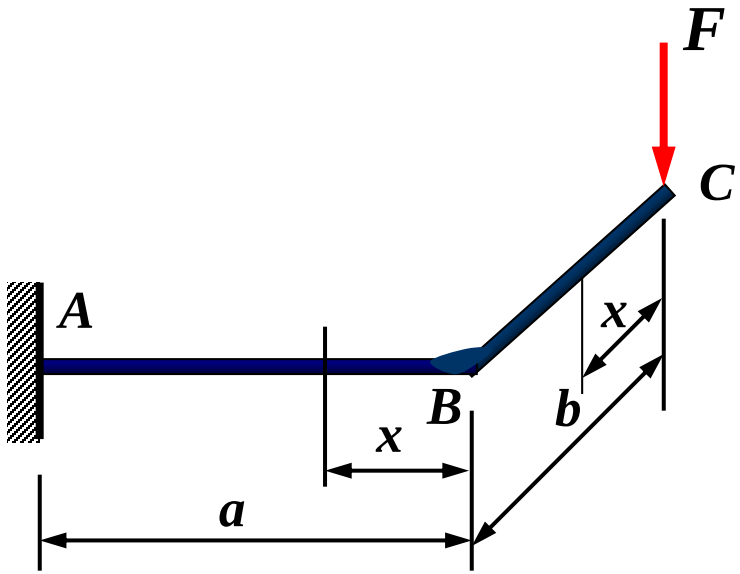


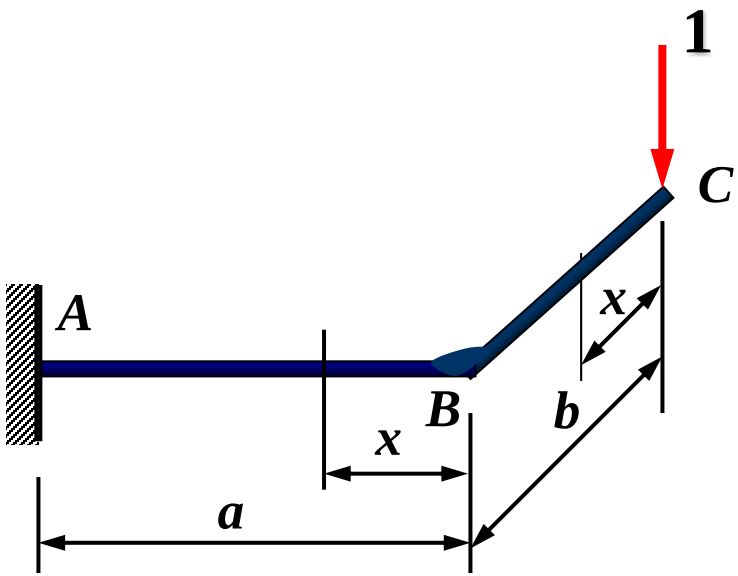




$$\boxed{\frac{F}{3EI}(a^3+b^3)+\frac{Fab^2}{GI_p}} \quad (\downarrow)$$







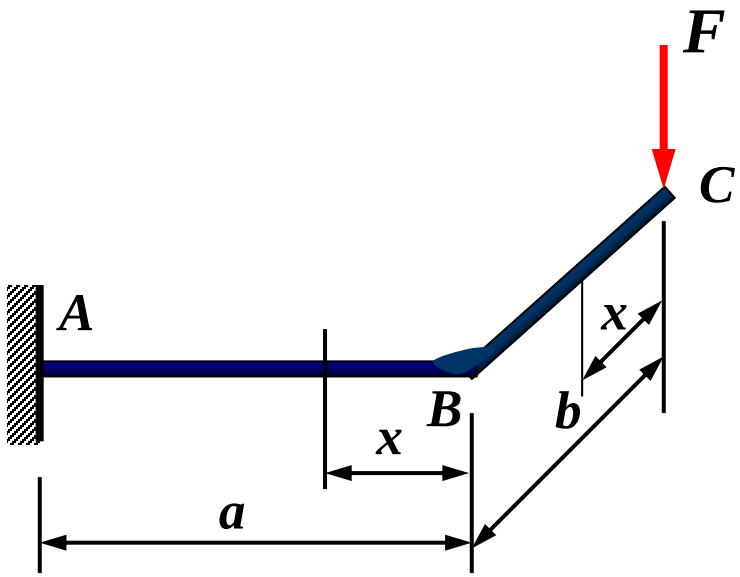
解:在 C点加竖向单位力

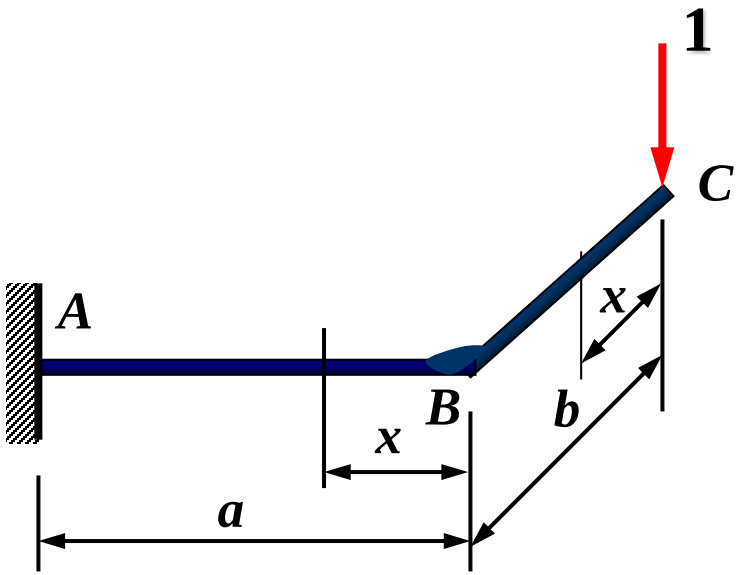
BC: $M(x) = -Fx$ $\quad \overline{M}(x) = -x$

$T(x) = 0$ $\quad \overline{T}(x) = 0$

AB: $M(x) = -Fx$ $\quad \overline{M}(x) = -x$

$T(x) = -Fb$ $\quad \overline{T}(x) = -b$





$$\Delta_C = \frac{1}{EI}\int_l M(x)\bar{M}(x)\mathrm{d}x + \frac{1}{GI_p}\int_l T(x)\bar{T}(x)\mathrm{d}x$$

$$= \frac{1}{EI}\int_0^a (-Fx)(-x)\mathrm{d}x + \frac{1}{EI}\int_0^b (-Fx)(-x)\mathrm{d}x +$$

$$\frac{1}{GI_p}\int_0^a (-Fb)(-b)\mathrm{d}x = \frac{F}{3EI}(a^3 + b^3) + \frac{Fab^2}{GI_p}(\downarrow)$$

作业

13.7-(b)（用单位载荷法及图乘法两种方法求解）
13.16

§ 13-8 计算莫尔积分的图乘法

(The method of moment areas for the mohr's integration)

等直杆的情况下，莫尔积分中的EI为常量，可提到积分号外面

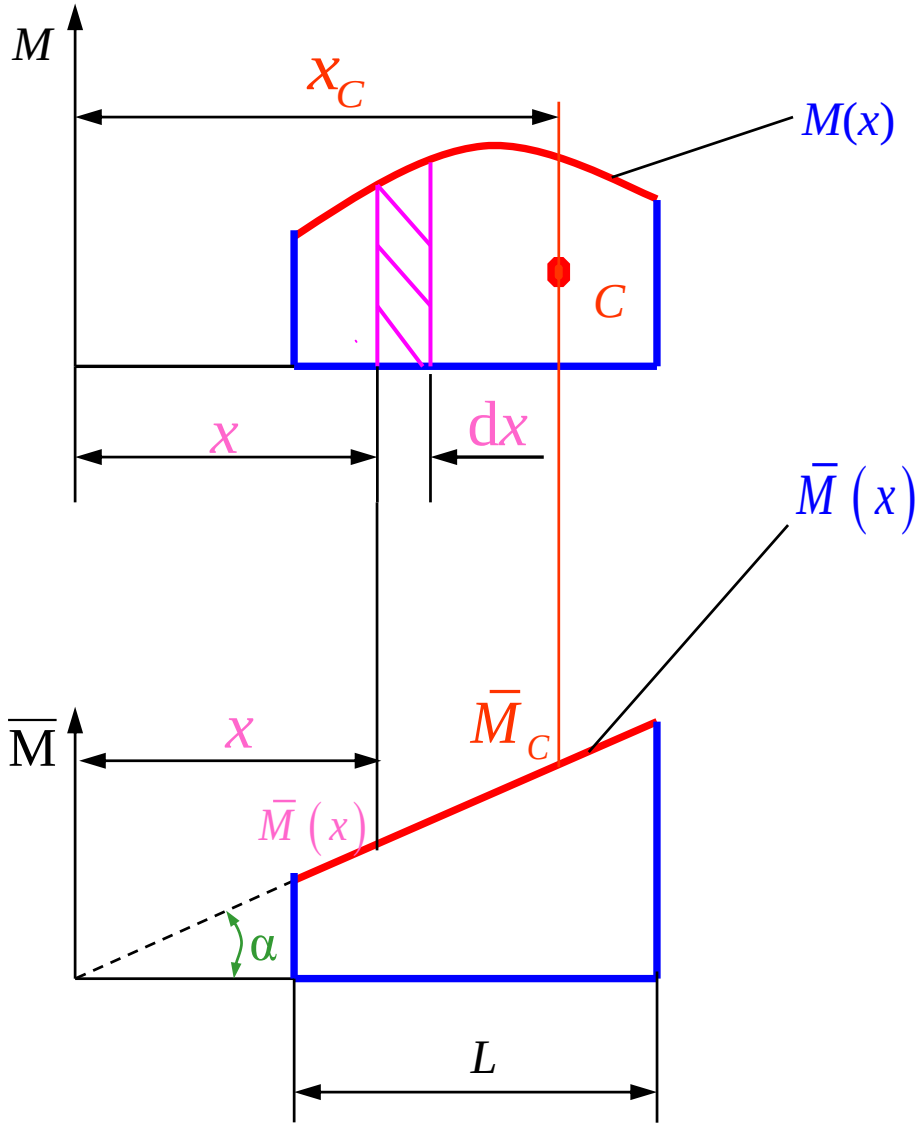
$$\Delta = \frac{1}{EI}\int_l M(x)\cdot\overline{M}(x)\mathrm{d}x$$

只需计算:

$$\int_l M(x)\cdot\overline{M}(x)\mathrm{d}x$$

因为 $\overline{M}(x)$ 是由单位力或单位力偶引起的弯矩，故沿杆长方向的 $\overline{M}(x)$ 图一般是由直线或折线组成。

$M(x)$图一般是曲线。



$$\int_l M(x)\cdot\overline{M}(x)\mathrm{d}x$$

$$=\tan\alpha\cdot\int_l x\cdot M(x)\mathrm{d}x$$

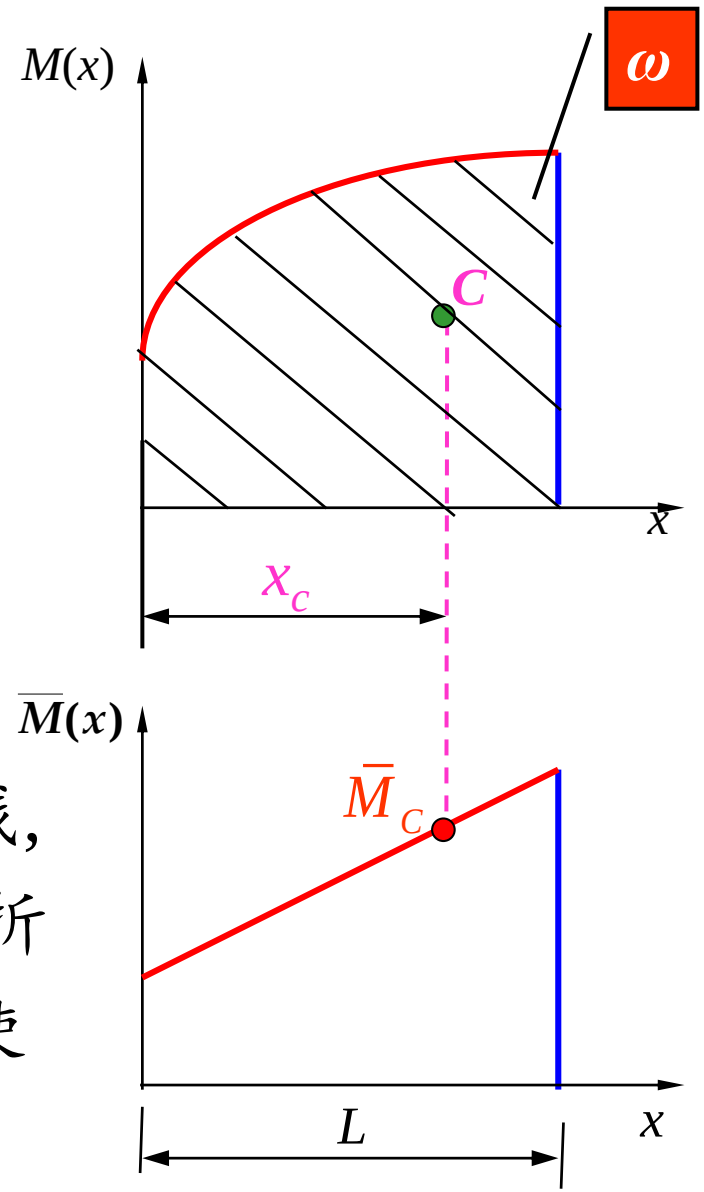
$$=\tan\alpha\cdot\omega\cdot x_C=\omega\cdot\overline{M}_C$$

$$\Delta=\int_l\frac{M(x)\overline{M}(x)}{EI}\mathrm{d}x=\frac{\omega\cdot\overline{M}_C}{EI}$$

$$\Delta = \frac{\omega \cdot \overline{M}_c}{EI}$$

$$\Delta = \frac{\bar{\omega} \cdot M_C}{EI}$$

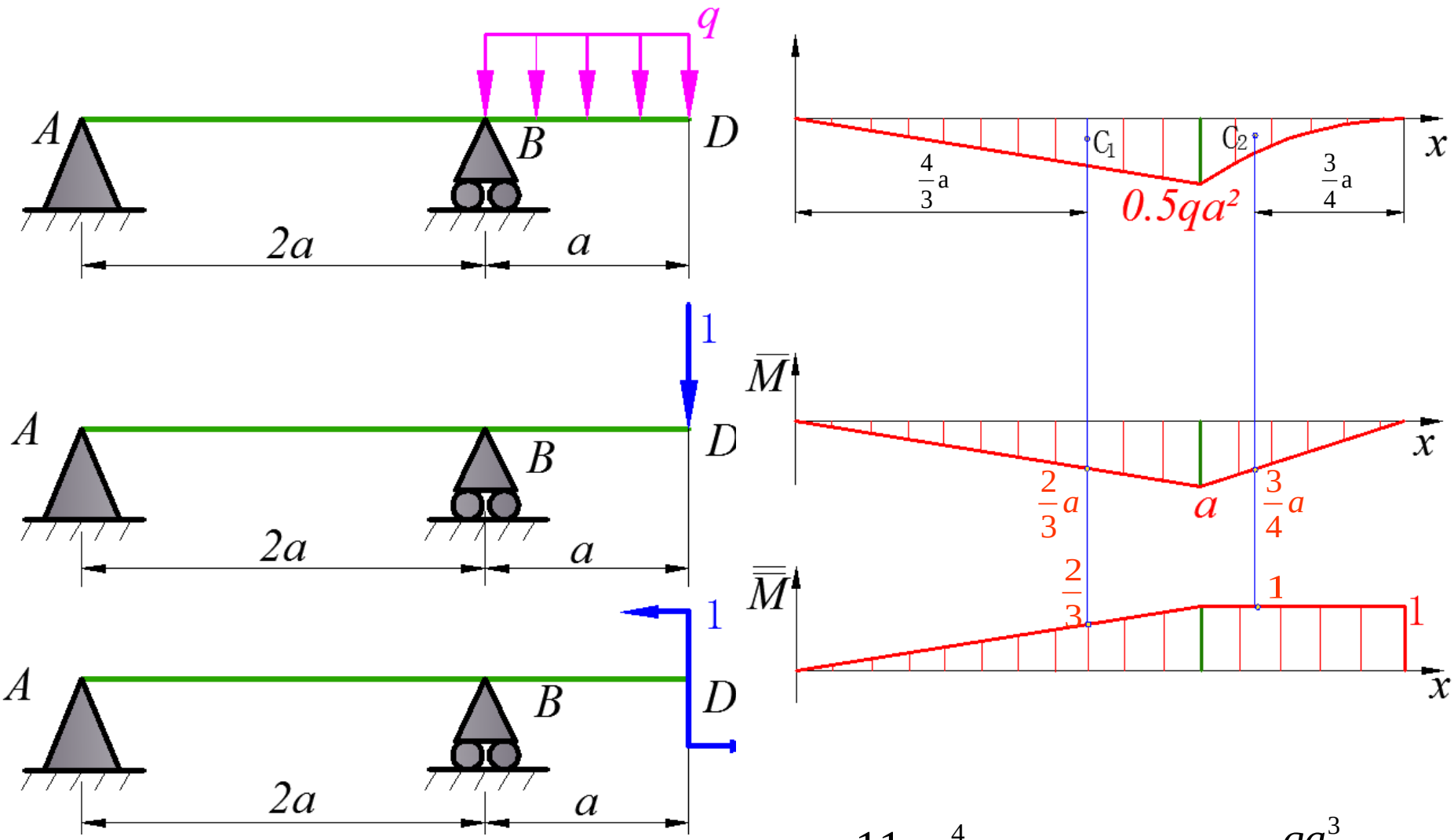
即积分可用$M(x)$图的面积ω和与$M(x)$图形心C对应的$\overline{M}_C$的乘积来代替。



注意：有时$M(x)$图为连续光滑曲线，而$\overline{M}(x)$为折线，则应以折线的转折点为界，把积分分成几段，逐段使用图乘法，然后求其和。

例题 13-5

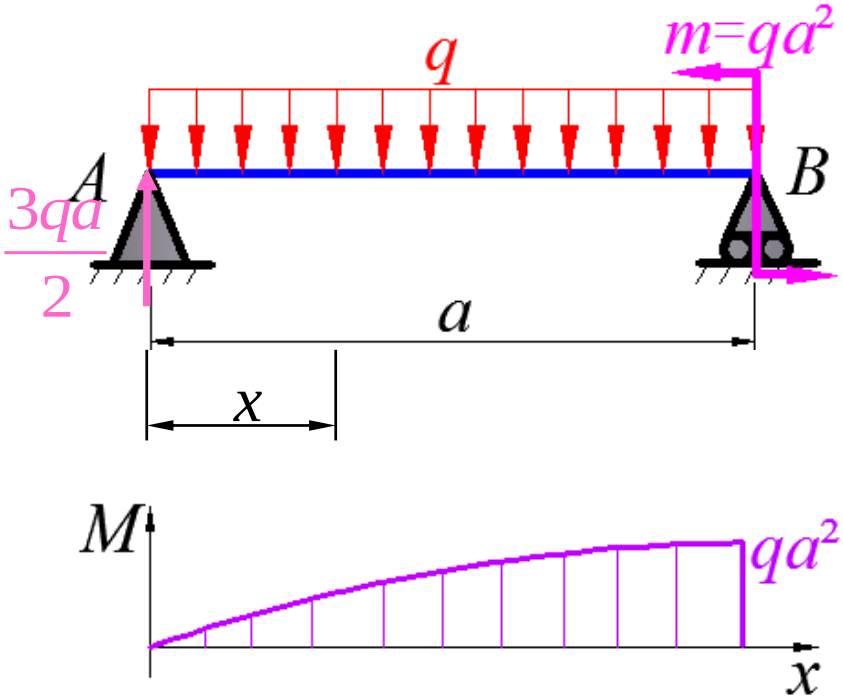
已知EI、q、a，求D截面的挠度和转角。

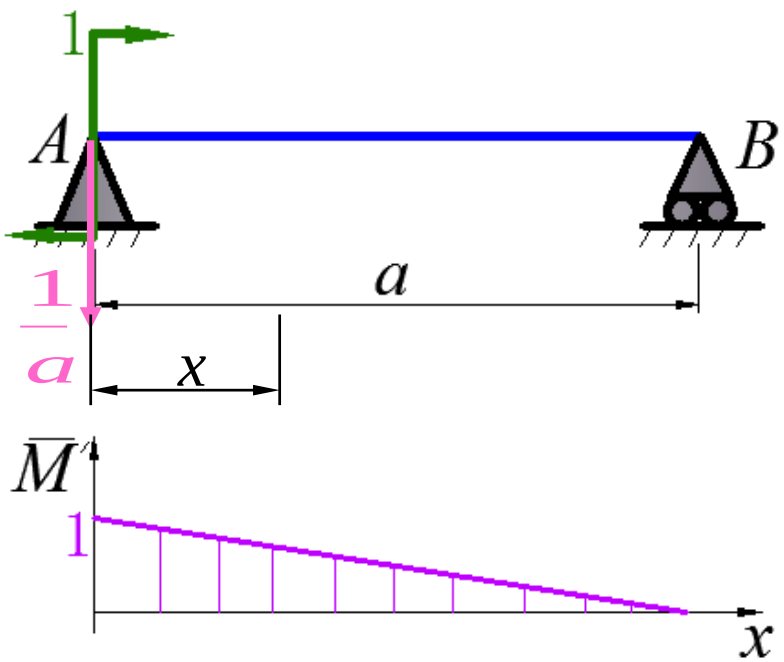


$$w_D = \frac{11qa^4}{24EI}(\downarrow) \qquad \theta_D = -\frac{qa^3}{2EI}(\circlearrowleft)$$

例题 13-6

已知EI、q、a，求A截面的转角。





$$M(x)=\frac{3qa}{2}x-\frac{q}{2}x^2$$

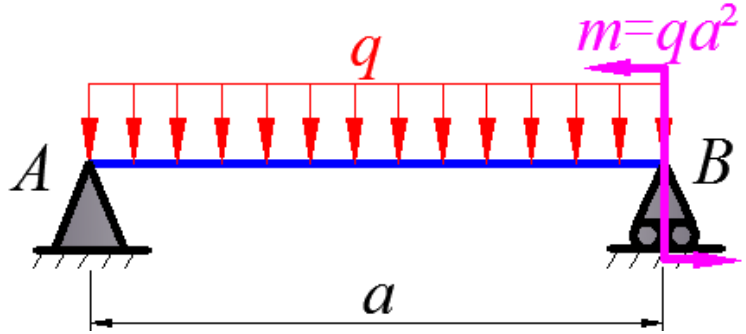
$$\bar{M}(x)=1-\frac{1}{a}x$$

$$\theta_A=\frac{5qa^3}{24EI}(\circlearrowleft)$$

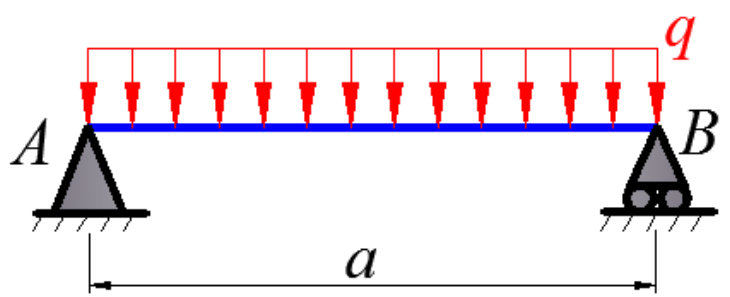
$$S_M=\int_0^a x\left(\frac{3qa}{2}x-\frac{q}{2}x^2\right)\mathrm{d}x$$

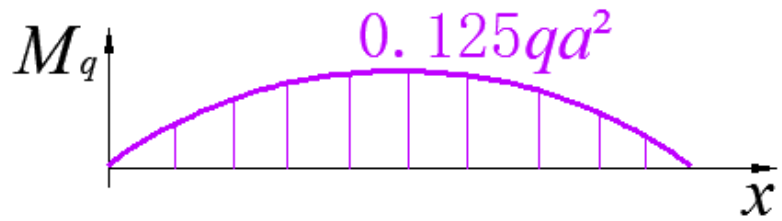
$$A=\int_0^a\left(\frac{3qa}{2}x-\frac{q}{2}x^2\right)\mathrm{d}x$$

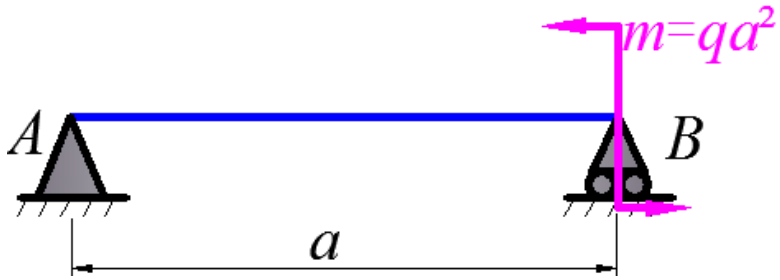
$$x_c=\frac{S_M}{A}$$

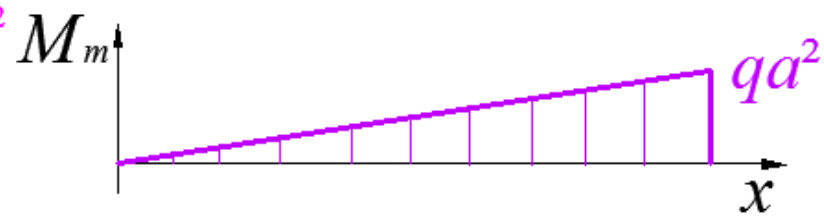


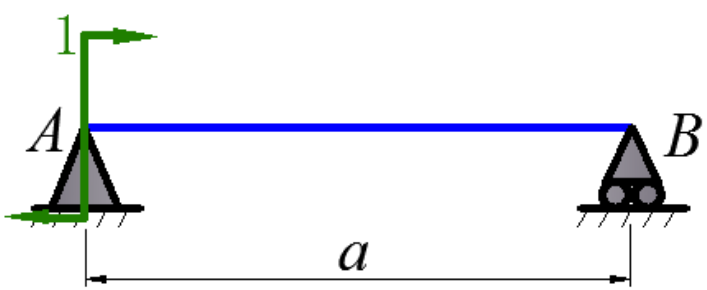
$$\theta_A = \frac{5qa^3}{24EI}(\lrcorner)$$

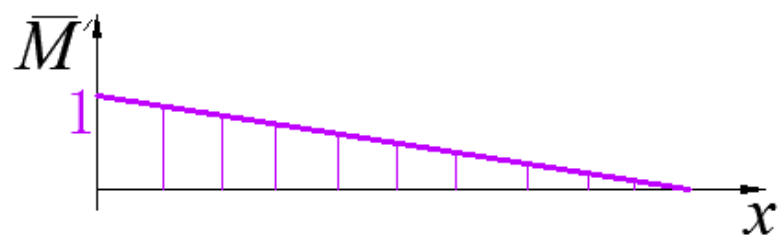






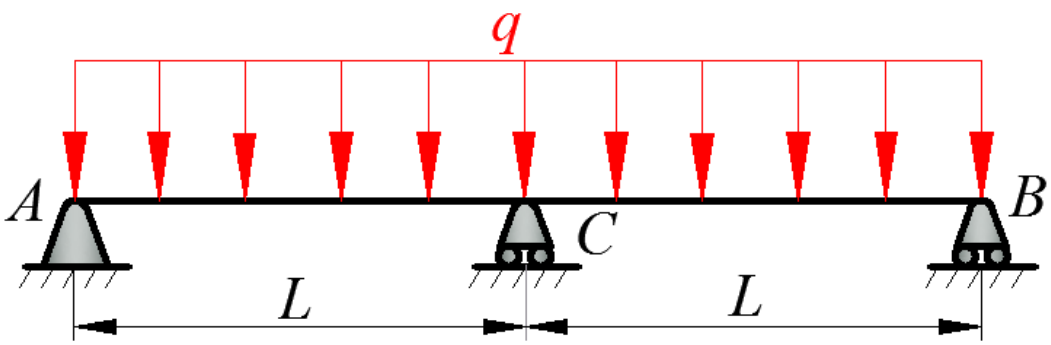




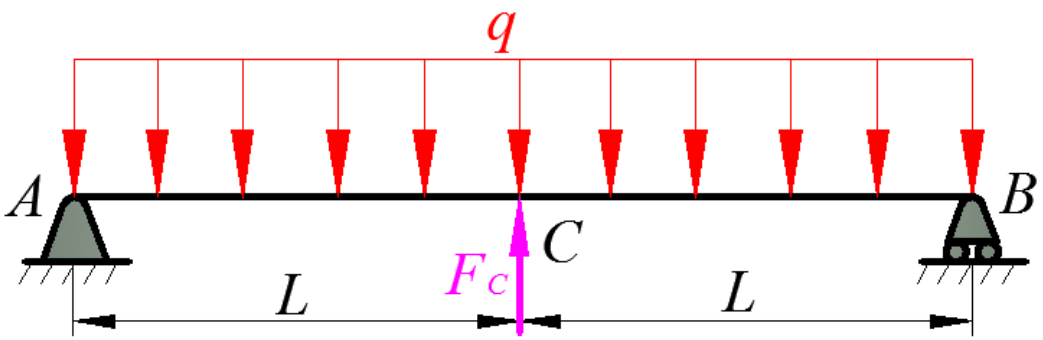


例题 13-7

如图所示等直梁，已知q、L。作出梁的剪力图和弯矩图。

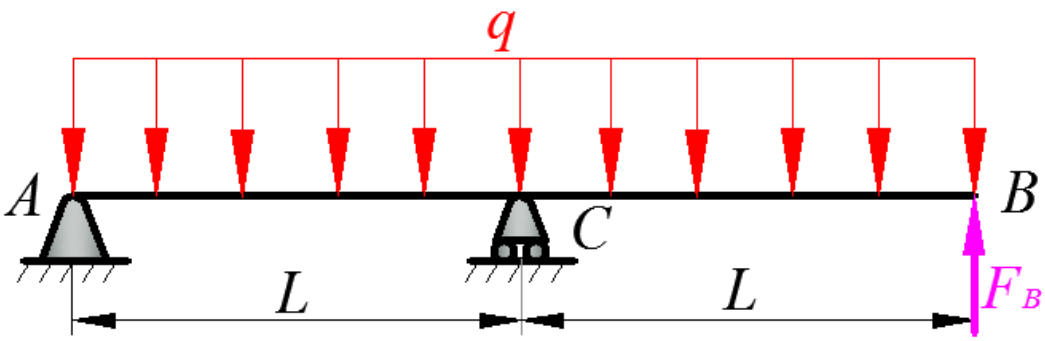


学号为单数者做。求F_c。

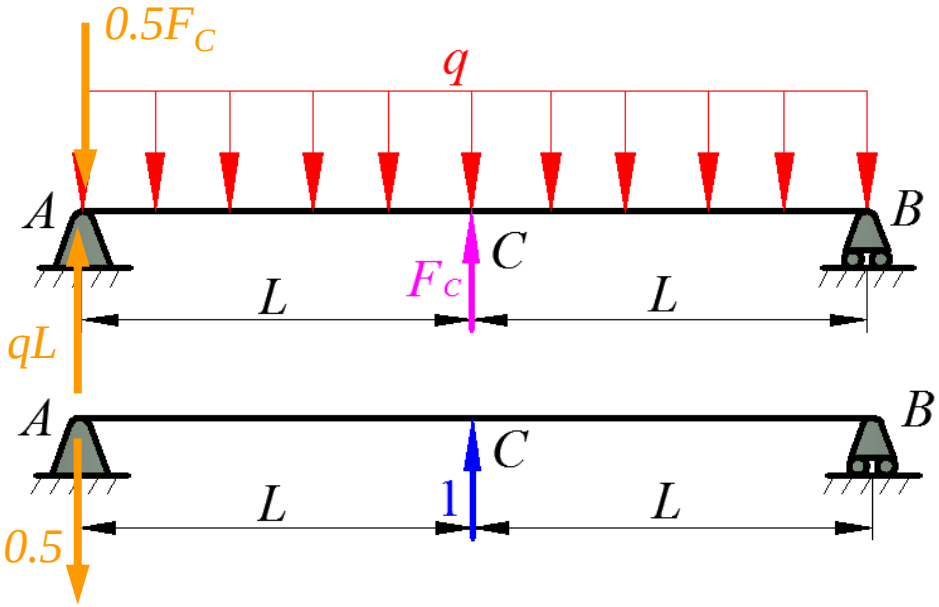


$w_C = 0$

学号为双数者做。求F_B。



$w_B = 0$



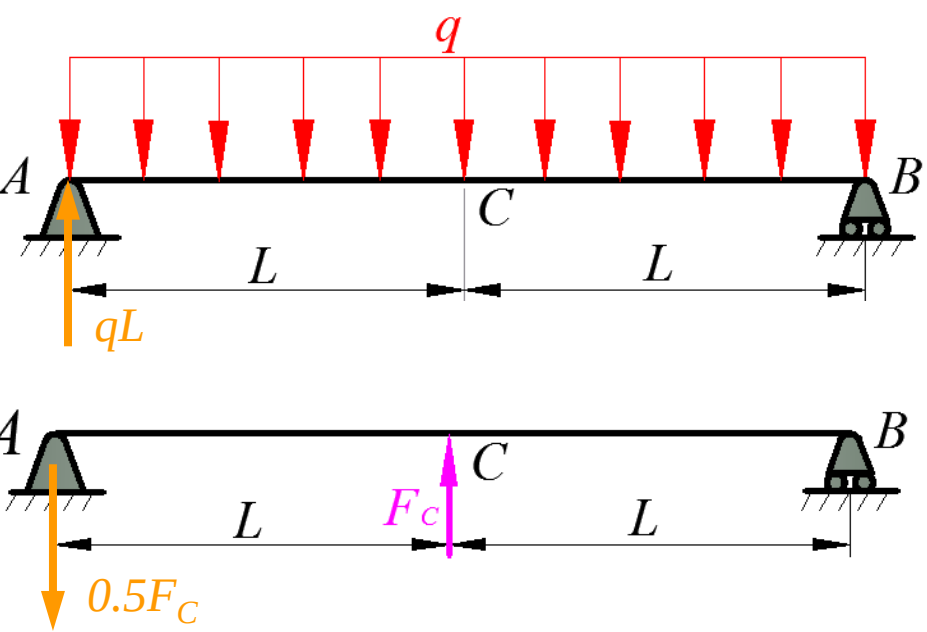
$$w_C = 0$$

$$M = qLx - 0.5F_C - 0.5qx^2$$

$$\bar{M} = -0.5x$$

$$w_C = \frac{2}{EI}\int_0^L M\bar{M}\,\mathrm{d}x = 0$$

$$F_C = \frac{5qL}{4}(\uparrow)$$



$$M_q = qLx - 0.5qx^2$$

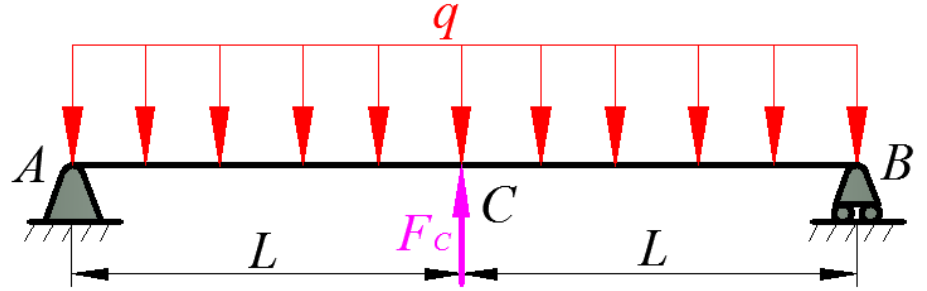
$$M_{F_C} = -0.5F_C x$$

$$\bar{M} = -0.5x$$

$$(w_C)_q = \frac{2}{EI}\int_0^L M_q\bar{M}\,\mathrm{d}x$$

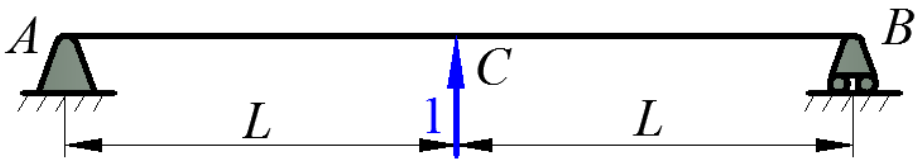
$$(w_C)_{F_C} = \frac{2}{EI}\int_0^L M_{F_C}\bar{M}\,\mathrm{d}x$$

$$(w_C)_q + (w_C)_{F_C} = 0$$

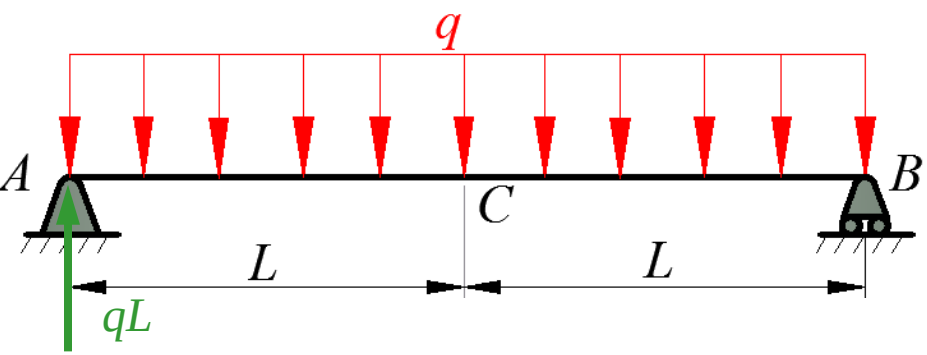


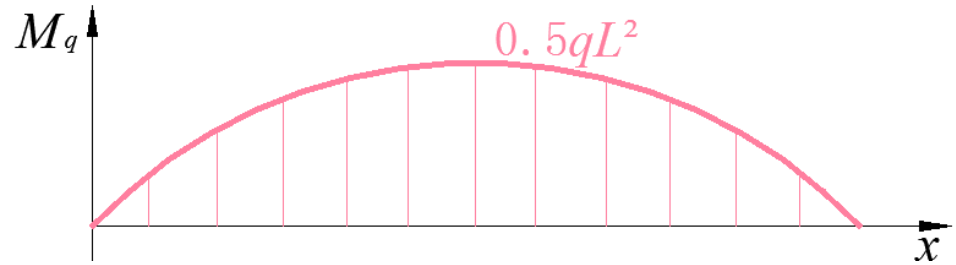
$$w_C = 0$$

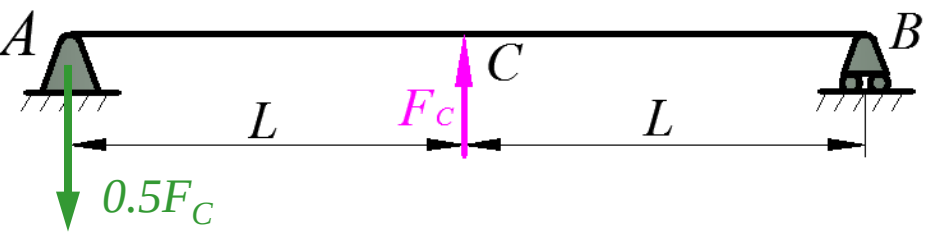
$$(w_C)_q + (w_C)_{F_C} = 0$$

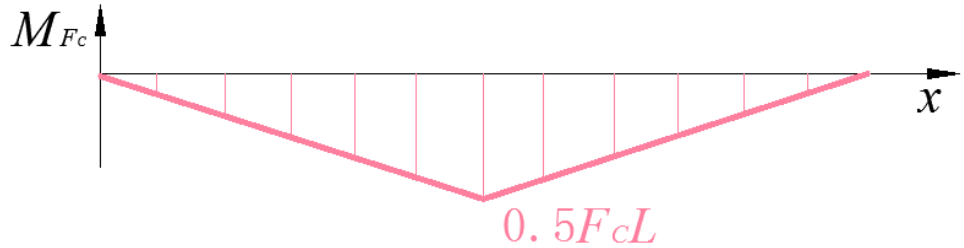


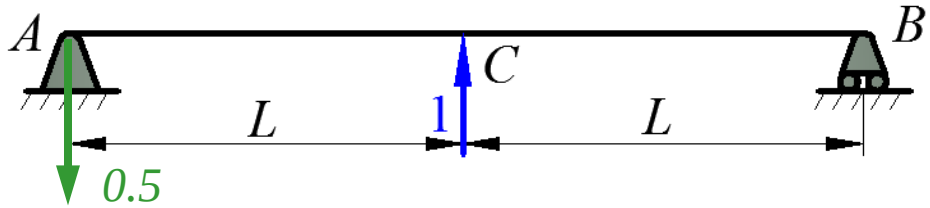
$$F_C = \frac{5qL}{4}(\uparrow)$$

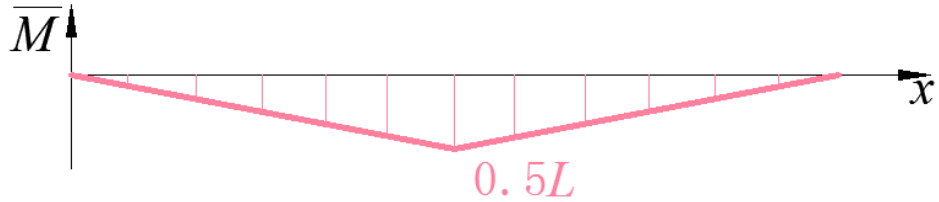


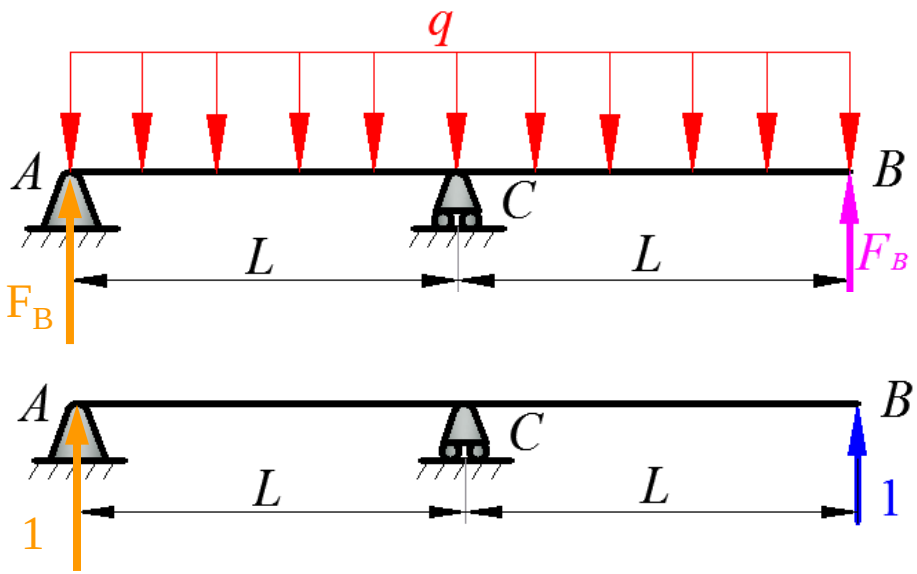












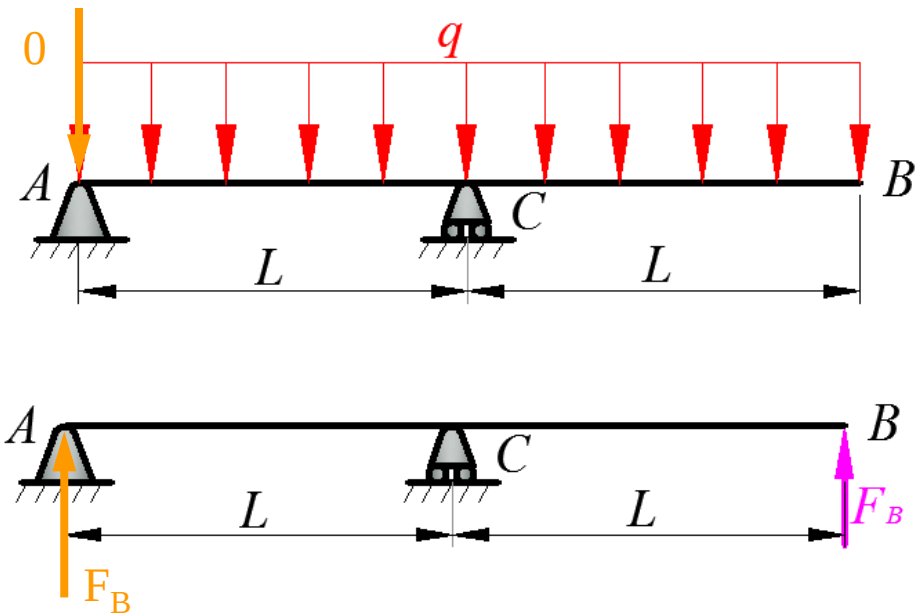
$$w_B = 0$$

$$M = F_B x - 0.5qx^2$$

$$\bar{M} = x$$

$$w_B = \frac{2}{EI}\int_0^L M\bar{M}\,\mathrm{d}x = 0$$

$$F_B = \frac{3qL}{8}(\uparrow)$$



$$M_q = -0.5qx^2$$

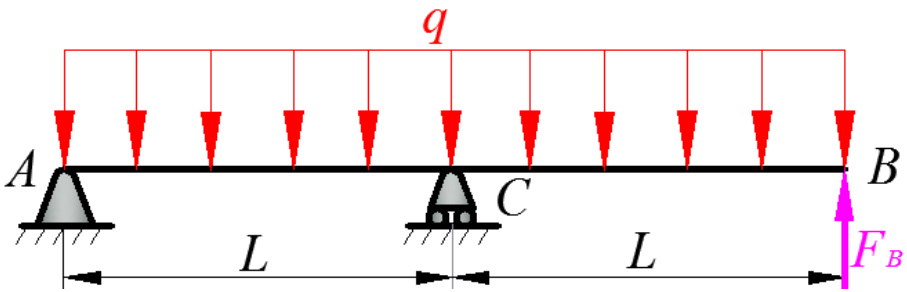
$$M_{F_C} = F_B x$$

$$\bar{M} = x$$

$$(w_B)_q = \frac{2}{EI}\int_0^L M_q\bar{M}\,\mathrm{d}x$$

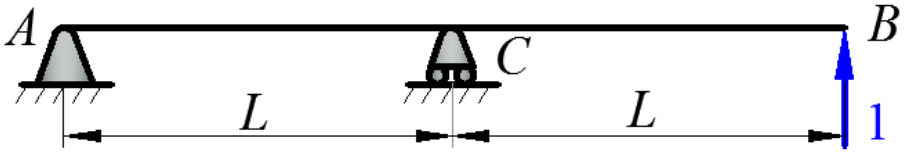
$$(w_B)_{F_B} = \frac{2}{EI}\int_0^L M_{F_B}\bar{M}\,\mathrm{d}x$$

$$(w_B)_q + (w_B)_{F_B} = 0$$

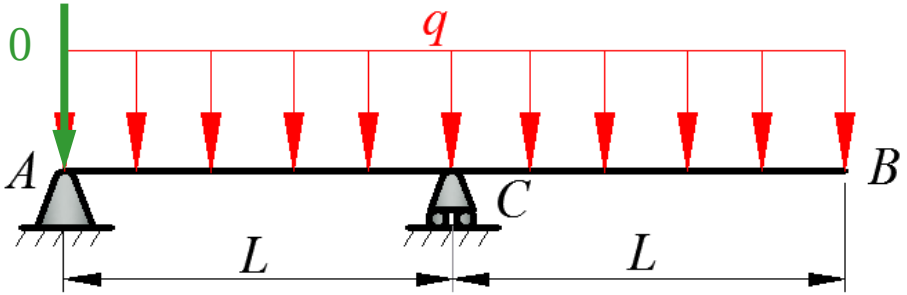


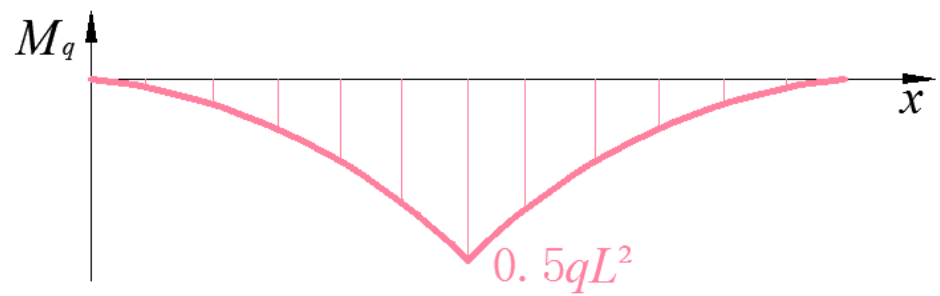
$$w_B = 0$$

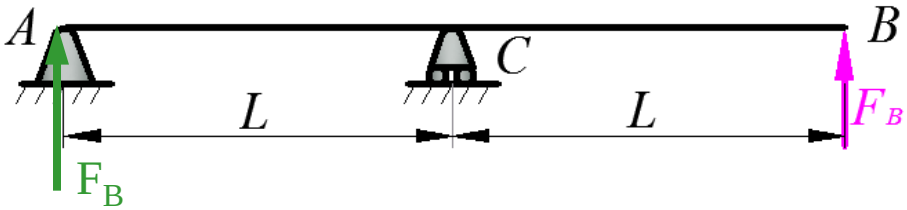
$$\left(w_B\right)_q + \left(w_B\right)_{F_B} = 0$$

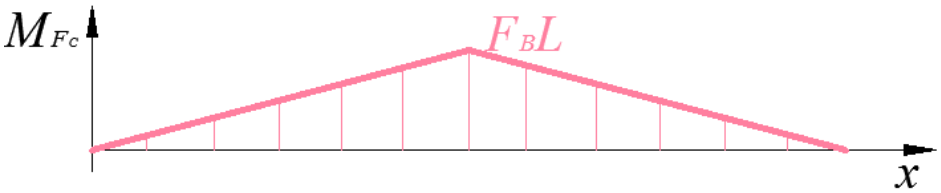


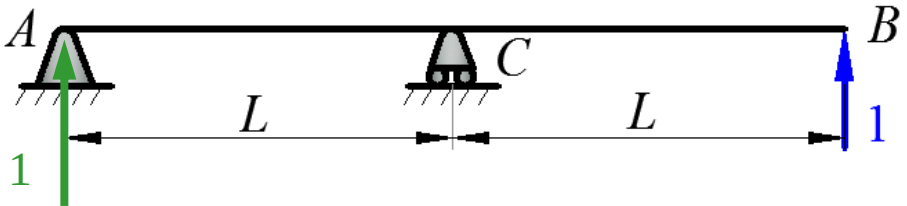
$$F_B = \frac{3qL}{8}(\uparrow)$$

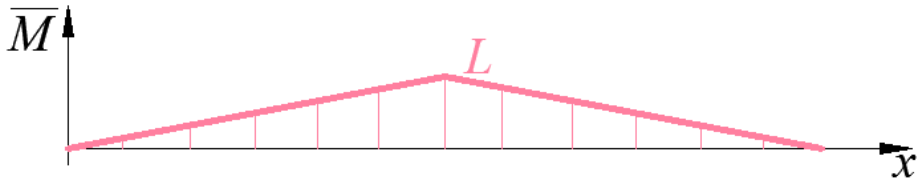


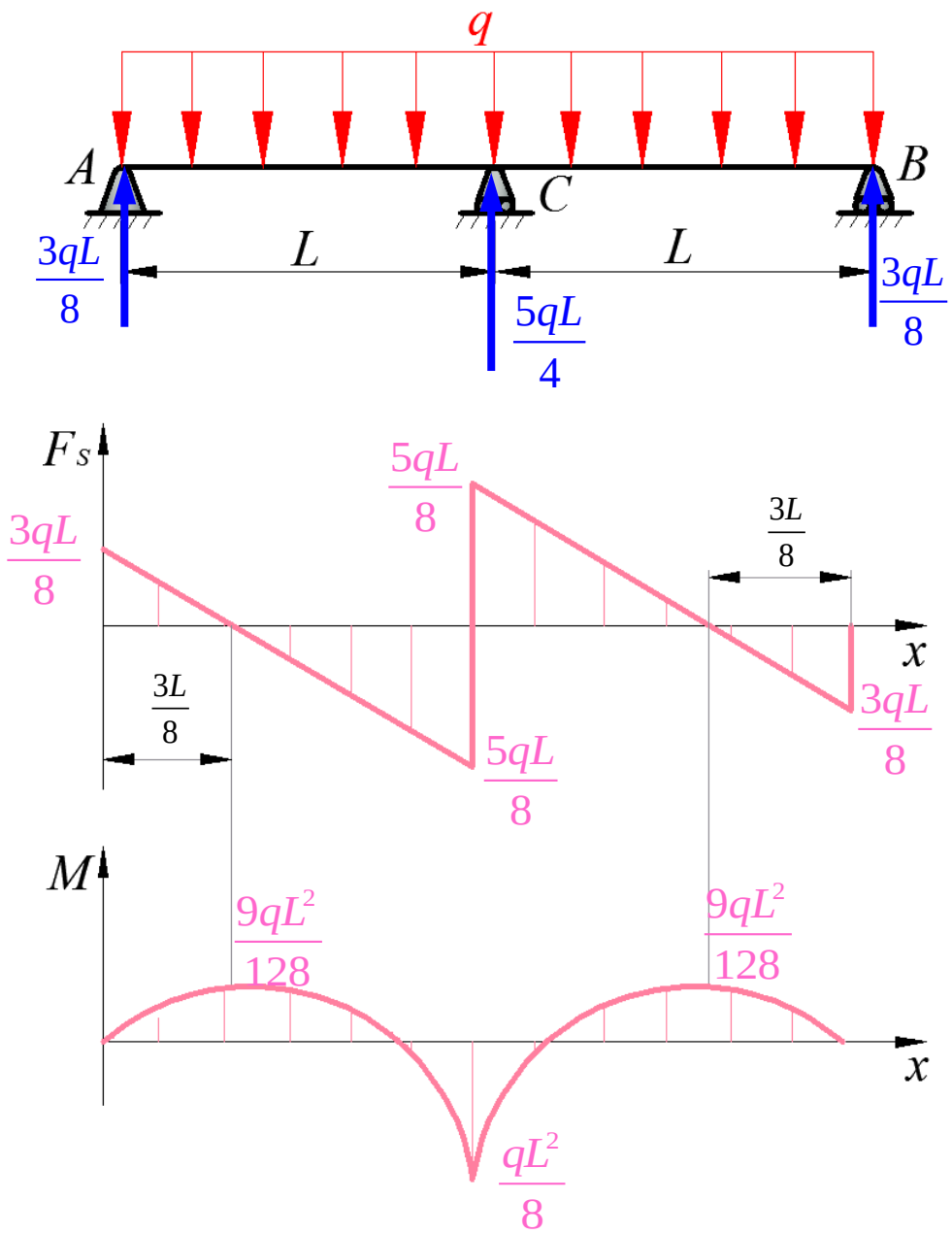











q
A
B
C
L
L
3qL/8
5qL/4
3qL/8
Fs
5qL/8
3qL/8
3L/8
x
3L/8
3qL/8
5qL/8
M
9qL²/128
9qL²/128
x
qL²/8

		拉压	剪切	扭转	弯曲
特点	外力特点				
	变形特点				
	横截面变化特点				
内力	名称				
	方位				
	正负规定				
	内力图				
应力	名称				
	横截面上分布特点				
	计算公式				
	最大应力计算公式				
变形	度量量				
	计算公式				

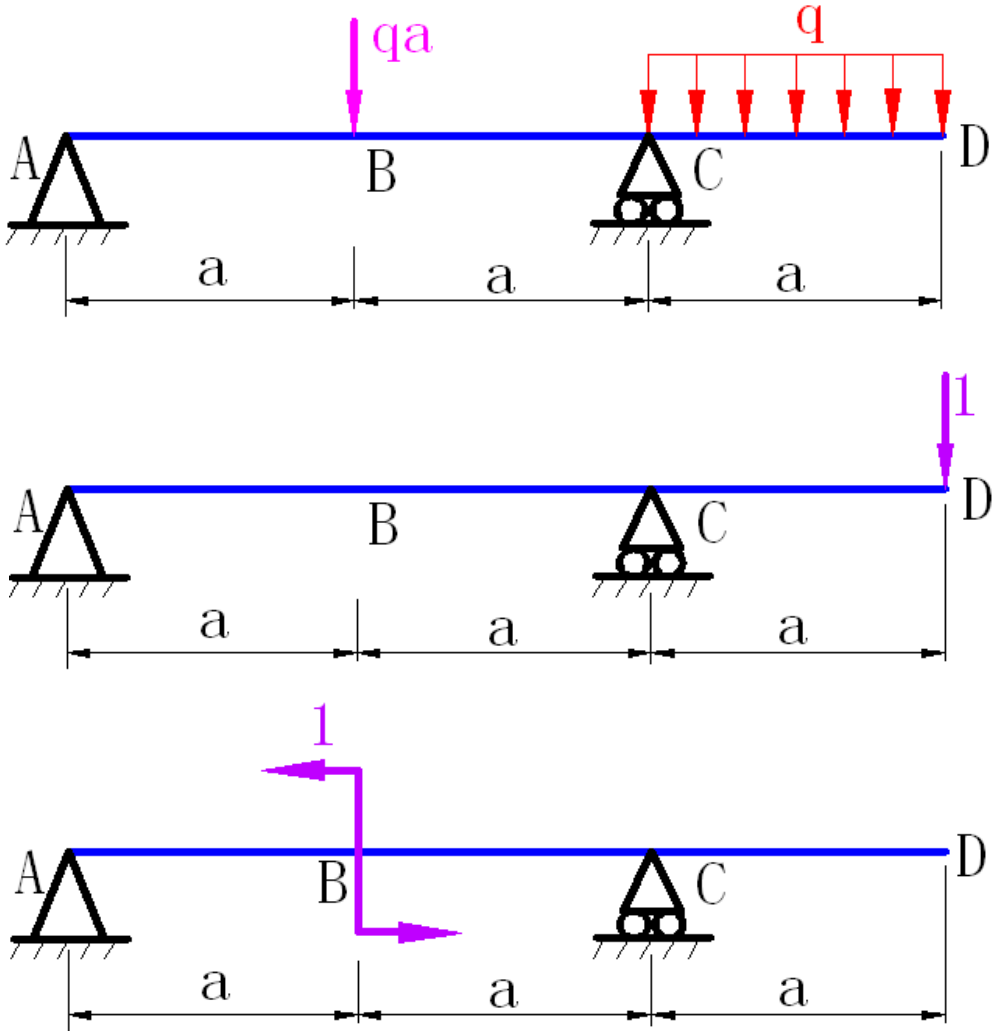
1、概念：强度、刚度、内力、应力、变形、位移、中性层、中性轴

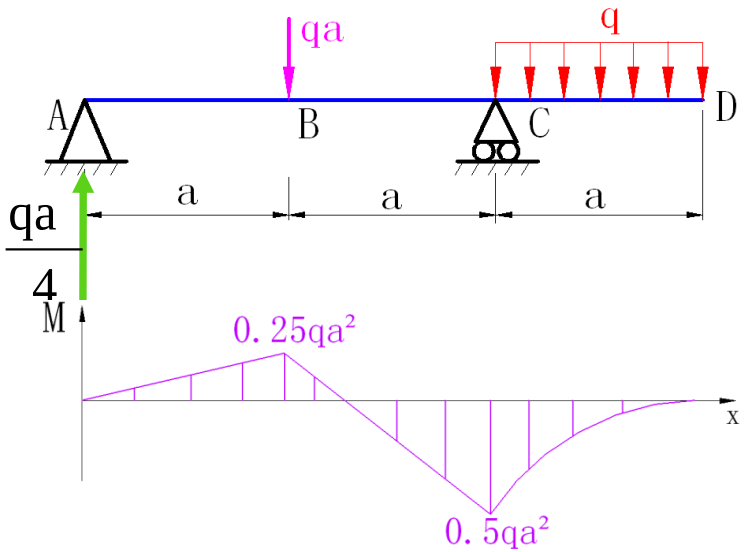
2、拉压、扭转实验：实验曲线、实验现象、破坏规律、指标值。

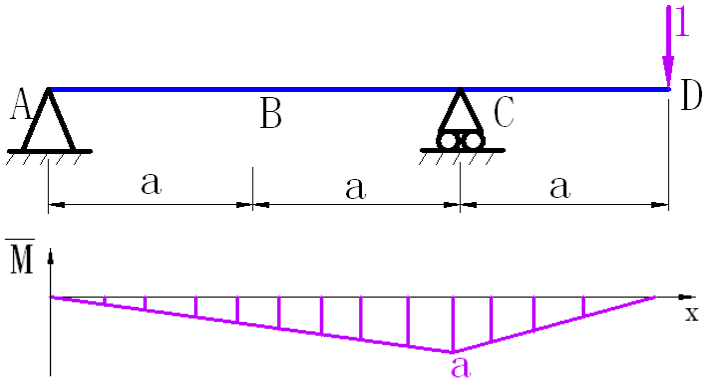
3、平面图形几何特性：静矩、形心、惯性矩、极惯性矩。

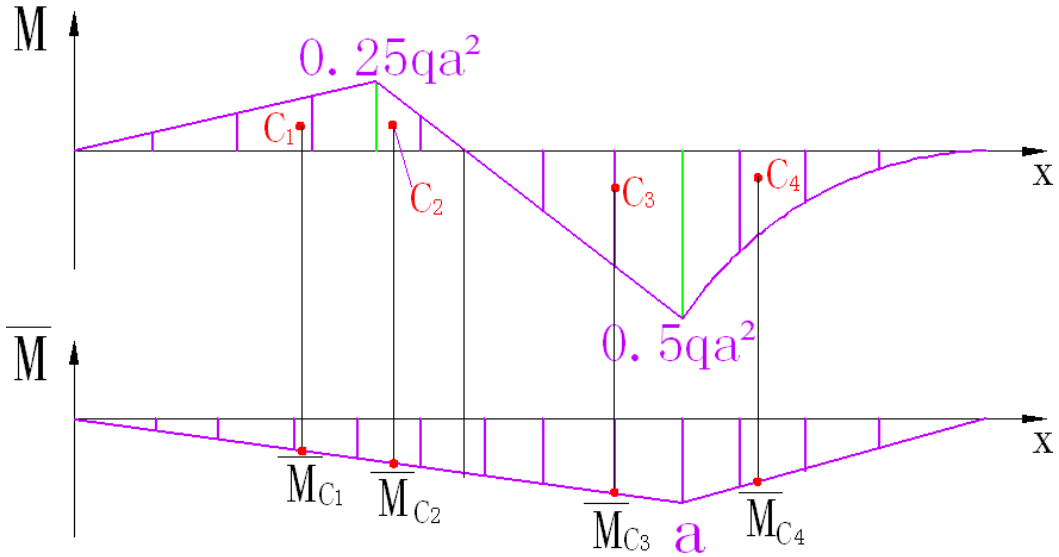
第十三章结束

例题13-5　已知EI、q、a，求D截面的挠度和B截面的转角。

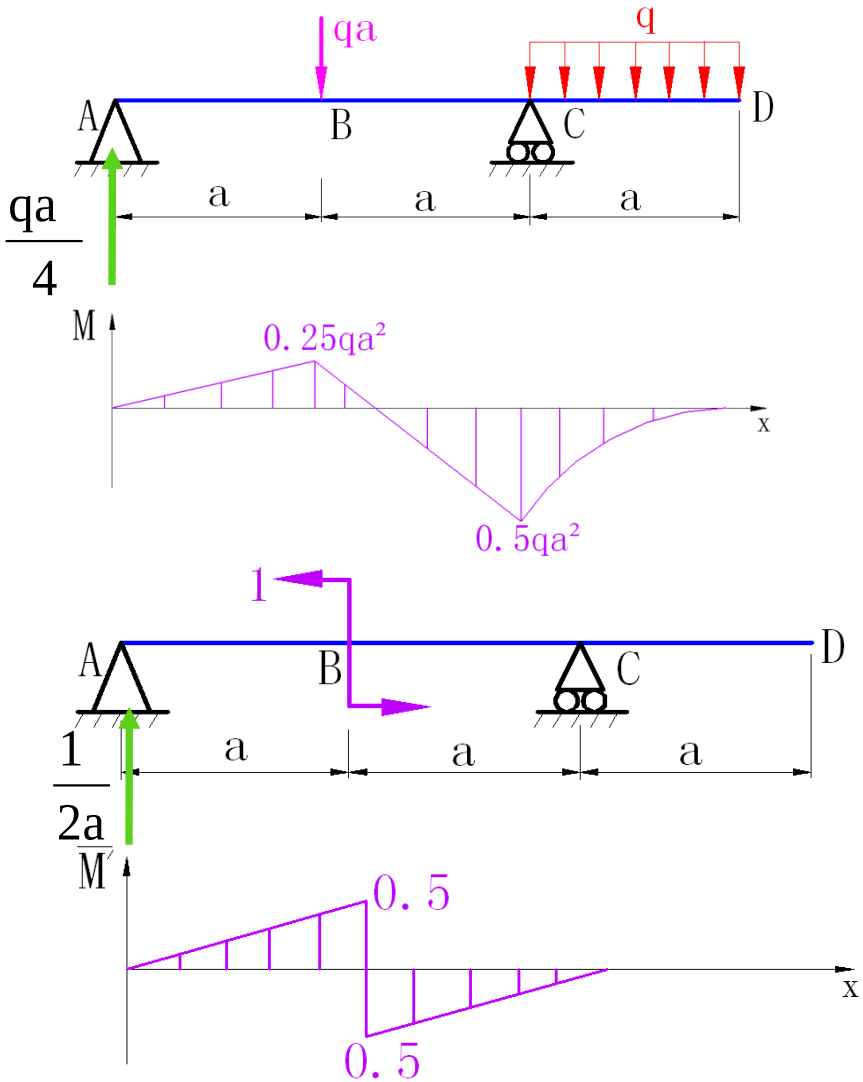


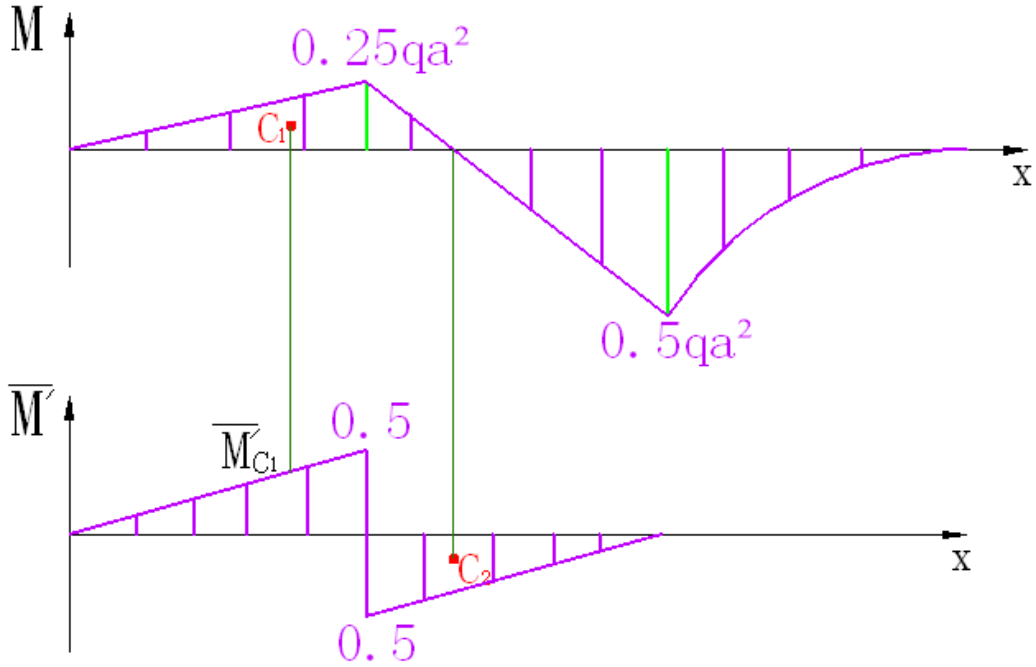






$$w_D = \frac{5qa^4}{24EI}(\downarrow)$$





$$\theta_B = \frac{qa^3}{24EI} \quad （逆）$$